

# TP, TPD

Installation and operating instructions



# TP, TPD

---

|                                                  |           |
|--------------------------------------------------|-----------|
| <b>English (GB)</b>                              |           |
| Installation and operating instructions. . . . . | 3         |
| <b>中文 (CN)</b>                                   |           |
| 安装和使用说明书 . . . . .                               | 17        |
| <b>Appendix . . . . .</b>                        | <b>31</b> |
| Declaration of conformity . . . . .              | 47        |

# English (GB) Installation and operating instructions

## Original installation and operating instructions

### CONTENTS

|                                         | Page      |
|-----------------------------------------|-----------|
| <b>1. Symbols used in this document</b> | <b>3</b>  |
| <b>2. General information</b>           | <b>3</b>  |
| <b>3. Delivery and handling</b>         | <b>4</b>  |
| 3.1 Delivery                            | 4         |
| 3.2 Handling                            | 4         |
| <b>4. Applications</b>                  | <b>5</b>  |
| 4.1 Pumped liquids                      | 5         |
| <b>5. Installation</b>                  | <b>5</b>  |
| 5.1 Pipework                            | 7         |
| 5.2 Elimination of noise and vibrations | 7         |
| 5.3 Foundation                          | 8         |
| 5.4 Terminal box positions              | 9         |
| 5.5 Base plate                          | 9         |
| 5.6 Insulation                          | 9         |
| 5.7 Frost protection                    | 9         |
| <b>6. Electrical connection</b>         | <b>10</b> |
| 6.1 Frequency converter operation       | 10        |
| <b>7. Startup</b>                       | <b>11</b> |
| 7.1 Flushing the pipe system            | 11        |
| 7.2 Priming                             | 11        |
| 7.3 Checking the direction of rotation  | 11        |
| 7.4 Starting                            | 12        |
| 7.5 Shaft seal run-in                   | 12        |
| 7.6 Frequency of starts and stops       | 12        |
| <b>8. Maintenance and service</b>       | <b>12</b> |
| 8.1 Pump                                | 12        |
| 8.2 Motor                               | 13        |
| 8.3 Service                             | 13        |
| 8.4 Adjusting the shaft                 | 13        |
| 8.5 Blanking flanges                    | 14        |
| <b>9. Technical data</b>                | <b>14</b> |
| 9.1 Ambient temperature                 | 14        |
| 9.2 Liquid temperature                  | 14        |
| 9.3 Operating pressure/test pressure    | 14        |
| 9.4 Inlet pressure                      | 14        |
| 9.5 Enclosure class                     | 14        |
| 9.6 Electrical data                     | 14        |
| 9.7 Sound pressure level                | 14        |
| 9.8 Environment                         | 14        |
| <b>10. Fault finding the product</b>    | <b>15</b> |
| <b>11. Disposal</b>                     | <b>16</b> |

#### Warning



Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

## 1. Symbols used in this document

#### Warning



If these safety instructions are not observed, it may result in personal injury.



If these safety instructions are not observed, it may result in malfunction or damage to the equipment.



Notes or instructions that make the job easier and ensure safe operation.

## 2. General information

These instructions apply to the pump types TP and TPD fitted with Grundfos motors. If the pump is fitted with another motor make, please note that the motor data may differ from the data stated in these instructions.

### 3. Delivery and handling

#### 3.1 Delivery

The pump is delivered from the factory in a carton with a wooden bottom, which is specially designed for transport by fork-lift truck or a similar vehicle.

#### 3.2 Handling

**Warning**

The lifting eyes fitted to large pump motors can be used for lifting the pump head (motor, motor stool and impeller). The lifting eyes must not be used for lifting the entire pump.



**TPD:**

The centrally positioned thread of the pump housing must not be used for lifting purposes as the thread is placed below the centre of gravity of the pump.

Pumps without lifting eyes must be lifted by means of nylon straps. See figures 1 and 2.

Pumps with lifting eyes must be lifted by means of nylon straps and shackles. See figures 3 and 4.

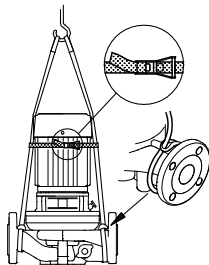


Fig. 1 TP

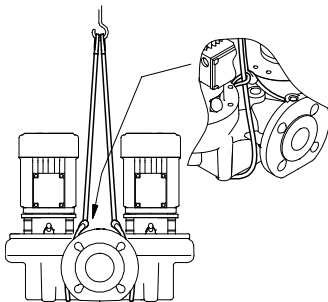


Fig. 2 TPD

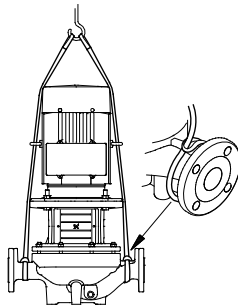


Fig. 3 TP

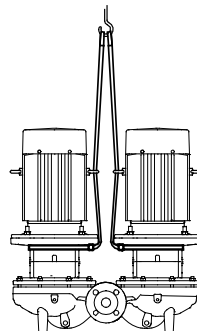


Fig. 4 TPD

TM02 7009 2303

TM02 7007 2303

TM02 7010 2303

TM02 7008 2303

## 4. Applications

The pumps are designed to circulate hot or cold water in residential, institutional and industrial applications in systems, such as:

- heating systems
- district heating plants
- central heating systems for blocks of flats
- air-conditioning systems
- cooling systems.

In addition, the pump range is used for liquid transfer and water supply in systems such as:

- washing systems
- domestic hot water systems
- industrial systems in general.

To ensure optimum operation, the dimensioning range of the system must fall within the performance range of the pump.

### 4.1 Pumped liquids

Thin, clean, non-aggressive and non-explosive liquids, not containing solid particles or fibres that may attack the pump mechanically or chemically.

Examples:

- Central heating system water (the water must meet the requirements of accepted standards on water quality in heating systems)
- cooling liquids
- domestic hot water
- industrial liquids
- softened water.

The pumping of liquids with a density and/or kinematic viscosity higher than that of water will have the following effects:

- a considerable pressure drop
- a drop in hydraulic performance
- a rise in power consumption.

In such cases, the pump must be fitted with a bigger motor. If in doubt, contact Grundfos.

The EPDM O-rings fitted as standard are primarily suitable for water.

If the water contains mineral/synthetic oils or chemicals or if other liquids than water are pumped, the O-rings must be chosen accordingly.

## 5. Installation

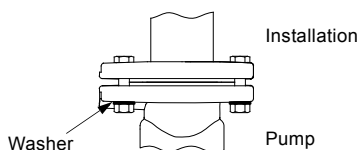


### Warning

When pumping hot or cold liquids, make sure that persons cannot accidentally come into contact with hot or cold surfaces.

The pump must be sited in a dry, well ventilated, but frost-free position.

When installing pumps with oval bolt holes in the pump flange (PN 6/10), use washers as shown in fig. 5.



**Fig. 5** Use of washers for oval bolt holes

Arrows on the pump housing show the direction of flow of liquid through the pump.

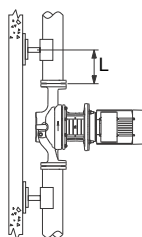
Pumps with motors smaller than 11 kW can be installed in horizontal or vertical pipework.

Pumps with motors of 11 kW and up may only be installed in horizontal pipework with the motor in vertical position.

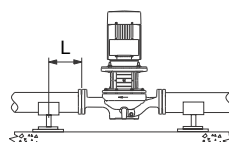
However, some TP, TPE pumps of 11 kW and up may be suspended directly in the pipes (horizontally or vertically). See the table [TP, TPE pumps from 11 kW and up suspended in the pipes](#) on page 43.

In installations where the pump is suspended directly in the pipes, the pump can support the pipe length L on both sides of the pump (L less than 3 x DN). See fig. 6. In installations where the pump is suspended directly in the pipes, the pump must be lifted and held in correct position by means of ropes or similar until both pump flanges are completely fastened to the pipe flanges.

### Vertical pipe



### Horizontal pipe



**Fig. 6** Pump suspended directly in the pipes

TM01 0683 1997

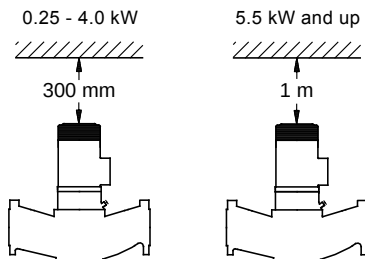
TM06 3518 0615

**Caution** The motor must never fall below the horizontal plane.

For inspection and motor/pump head removal, the following clearance is required above the motor:

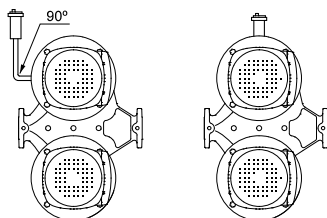
- 300 mm for motors up to and including 4.0 kW.
- 1 m for motors of 5.5 kW and up.

See fig. 7.



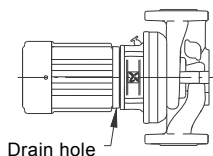
**Fig. 7** Required clearance above the motor

Twin-head pumps installed in horizontal pipes must be fitted with an automatic air vent in the upper part of the pump housing. See fig. 8. The automatic air vent is not supplied with the pump.



**Fig. 8** Automatic air vent

If the liquid temperature falls below the ambient temperature, condensation may form in the motor during inactivity. In this case, make sure that the drain hole in the motor flange is open and points downwards. See fig. 9.



**Fig. 9** Drain hole in motor flange

If twin-head pumps are used for pumping liquids with a temperature below 0 °C/32 °F, condensed water may freeze and cause the coupling to get stuck. The problem can be remedied by installing heating elements. Whenever possible (pumps with motors smaller than 11 kW), the pump must be installed with the motor shaft in horizontal position. See fig. 8.

**Caution** The technical data in section 9. *Technical data* must be observed.

## 5.1 Pipework

Fit isolating valves on either side of the pump to avoid draining the system if the pump needs to be cleaned or repaired.

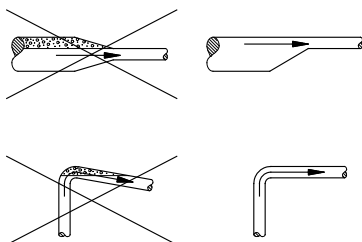
The pump is suitable for pipeline mounting, provided that the pipework is adequately supported either side of the pump. TP 25-50, 25-80, 25-90, 32-50, 32-80, 32-90, 40-50, 40-80 and 40-90 are designed for pipeline mounting only.

When installing the pipes, make sure that the pump housing is not stressed by the pipework.

The inlet and outlet pipes must be of an adequate size, taking the pump inlet pressure into account.

To avoid sediment build-up, do not fit the pump at the lowest point of the system.

Install the pipes so that air locks are avoided, especially on the inlet side of the pump. See fig. 10.



TM00 2263 0195

**Fig. 10** Correct pipework on the inlet side of the pump

### Caution

The pump is not allowed to run against a closed outlet valve as this will cause an increase in temperature/formation of steam in the pump which may cause damage to the pump.

If there is any danger of the pump running against a closed outlet valve, ensure a minimum liquid flow through the pump by connecting a bypass/a drain to the outlet pipe. The drain can for instance be connected to a tank. A minimum flow rate equal to 10 % of the flow rate at maximum efficiency is needed at all times.

Flow rate and head at maximum efficiency are stated on the pump nameplate.

## 5.2 Elimination of noise and vibrations

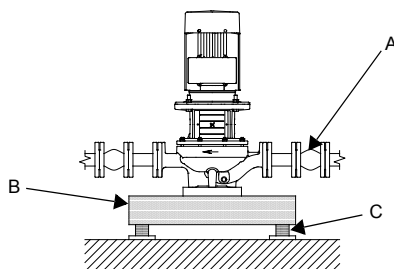
In order to achieve optimum operation and minimum noise and vibration, consider vibration damping of the pump. Generally, always consider this for pumps with motors of 11 kW and up, but for motors of 90 kW and up as well as the pumps stated in the table below, vibration damping is mandatory:

| Pump type    | P2 [kW] | Frequency [Hz] |
|--------------|---------|----------------|
| TP 200-280/4 | 37      | 60             |
| TP 200-290/4 | 37      | 50             |
| TP 200-320/4 | 45      | 60             |
| TP 200-360/4 | 55      | 60             |
| TP 200-390/4 | 75      | 60             |

Smaller motor sizes, however, may also cause undesirable noise and vibration.

Noise and vibration are generated by the revolutions of the motor and pump and by the flow in pipes and fittings. The effect on the environment is subjective and depends on correct installation and the state of the remaining system.

Elimination of noise and vibrations is best achieved by means of a concrete foundation, vibration dampers and expansion joints.



**Fig. 11** Foundation for TP pump

| Pos. | Description       |
|------|-------------------|
| A    | Expansion joint   |
| B    | Concrete pedestal |
| C    | Vibration damper  |

TM02 4993 3202

At high liquid velocities (greater than 5 m/s), we recommend that you fit larger expansion joints matching the pipes.

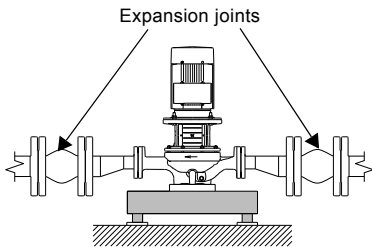


Fig. 12 TP pump installed with larger expansion joints

5.3 Foundation

We recommend that you install the pump on a concrete foundation which is heavy enough to provide permanent and rigid support to the entire pump. The foundation must be capable of absorbing any vibration, normal strain or shock. As a rule of thumb, the weight of the concrete foundation must be 1.5 times the weight of the pump. Place the pump on the foundation and fasten it. See fig. 11.

5.3.1 Recommended concrete foundations for TP, TPD Series 300 pumps

For TP Series 300 pumps with weights of 150 kg or more, we recommend that you mount the pump on a concrete foundation with the dimensions stated in the table below. The same recommendation applies for TPD Series 300 pumps with weights of 300 kg or more.

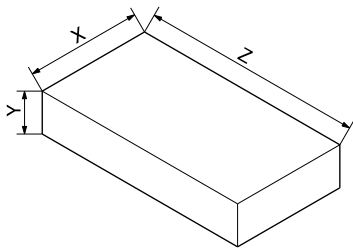


Fig. 13 Foundation for TP, TPD Series 300 pumps

| Concrete foundation dimensions |                 |                 |                |
|--------------------------------|-----------------|-----------------|----------------|
| Pump weight [kg]               | Y (height) [mm] | Z (length) [mm] | X (width) [mm] |
| 150                            | 280             | 565             | 565            |
| 200                            | 310             | 620             | 620            |
| 250                            | 330             | 670             | 670            |
| 300                            | 360             | 710             | 710            |
| 350                            | 375             | 750             | 750            |
| 400                            | 390             | 780             | 780            |
| 450                            | 410             | 810             | 810            |
| 500                            | 420             | 840             | 840            |
| 550                            | 440             | 870             | 870            |
| 600                            | 450             | 900             | 900            |
| 650                            | 460             | 920             | 920            |
| 700                            | 470             | 940             | 940            |
| 750                            | 480             | 970             | 970            |
| 800                            | 490             | 990             | 990            |
| 850                            | 500             | 1010            | 1010           |
| 900                            | 510             | 1030            | 1030           |
| 950                            | 520             | 1050            | 1050           |
| 1000                           | 530             | 1060            | 1060           |
| 1050                           | 540             | 1080            | 1080           |
| 1100                           | 550             | 1100            | 1100           |
| 1150                           | 560             | 1100            | 1100           |
| 1200                           | 560             | 1130            | 1130           |
| 1250                           | 570             | 1150            | 1150           |
| 1300                           | 580             | 1160            | 1160           |
| 1350                           | 590             | 1180            | 1180           |
| 1400                           | 600             | 1190            | 1190           |
| 1450                           | 600             | 1200            | 1200           |
| 1500                           | 610             | 1220            | 1220           |
| 1550                           | 620             | 1230            | 1230           |
| 1600                           | 620             | 1250            | 1250           |
| 1650                           | 630             | 1250            | 1250           |
| 1700                           | 635             | 1270            | 1270           |

TM04 9629 4810

TM03 9190 3607

≤ DN 200



Concrete foundation dimensions

| Pump weight<br>[kg] | Y<br>(height)<br>[mm]          | Z<br>(length)<br>[mm] | X<br>(width)<br>[mm] |
|---------------------|--------------------------------|-----------------------|----------------------|
| 800                 | 450                            | 1400                  | 800                  |
| 1000                | 450                            | 1400                  | 1000                 |
| 1200                | 450                            | 1400                  | 1200                 |
| 1400                | 500                            | 1600                  | 1200                 |
| 1600                | 500                            | 1600                  | 1350                 |
| 1800                | 500                            | 1600                  | 1500                 |
| 2000                | 550                            | 1600                  | 1600                 |
| 2200                | 550                            | 1700                  | 1700                 |
| 2400                | DN 300 /<br>DN 350 /<br>DN 400 | 550                   | 1800                 |
| 2600                |                                | 600                   | 1800                 |
| 3000                |                                | 600                   | 2000                 |
| 3400                |                                | 680                   | 2000                 |
| 3800                |                                | 760                   | 2000                 |
| 4200                |                                | 840                   | 2000                 |
| 4600                |                                | 920                   | 2000                 |
| 5000                |                                | 1000                  | 2000                 |
| 5400                |                                | 1080                  | 2000                 |

## 5.4 Terminal box positions



### Warning

Before starting work on the pump, make sure that the power supply has been switched off and that it cannot be accidentally switched on.

The terminal box can be turned to any of four positions, in 90 ° steps.

Change the terminal box position as follows:

1. If necessary, remove the coupling guards using a screwdriver. Do not remove the coupling.
2. Remove the screws securing the motor to the pump.
3. Turn the motor to the required position.
4. Replace and tighten the screws.
5. Replace the coupling guards.

## 5.5 Base plate

Single-head pumps (except TP 25-50, 25-80, 25-90, 32-50, 32-80, 32-90, 40-50, 40-80 and 40-90) have two tapped holes in the bottom of the pump housing which can be used for fitting a Grundfos base plate to the pump. The base plate is available as an optional extra.

Twin-head pumps have four tapped holes in the bottom of the pump housing. For some twin-head pumps, a base plate consisting of two halves is available.

Base plates with dimensions are shown on page 46.

## 5.6 Insulation

Do not insulate the motor stool as this will trap any vapour escaping from the shaft seal, thus causing corrosion. Covering the motor stool with insulation will also make inspection and service difficult.

Follow the guidelines in fig. 14 when insulating the pump.

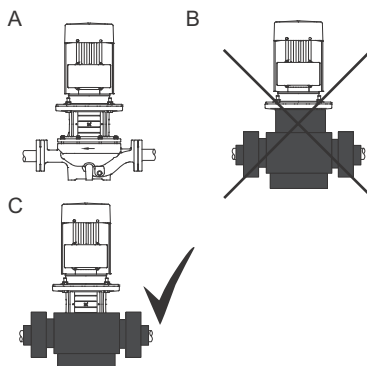


Fig. 14 Insulation of TP pumps

| Pos. | Description          |
|------|----------------------|
| A    | Without insulation   |
| B    | Incorrect insulation |
| C    | Correct insulation   |

## 5.7 Frost protection

Pumps which are not being used during periods of frost must be drained to avoid damage.

TM05 2328 4911

## 6. Electrical connection

Make the electrical connection in accordance with local regulations.

### Warning



Before removing the terminal box cover and before any removal/dismantling of the pump, make sure that the power supply has been switched off.

Connect the pump to an external mains switch with a minimum contact gap of 3 mm in all poles.

The operating voltage and frequency are marked on the pump nameplate. Make sure that the motor is suitable for the power supply on which it will be used.

Single-phase standard motors incorporate a thermal switch and require no additional motor protection.

Three-phase motors must be connected to a motor protection device.

Motors of 3 kW and up incorporate thermistors (PTC). The thermistors are designed according to DIN 44082.

Make the electrical connection as shown in the diagram inside the terminal box cover.

The motors of twin-head pumps are to be connected separately.

### Caution

Do not start the pump until it has been filled with liquid and vented.

## 6.1 Frequency converter operation

Motors types Siemens, MG 71 and MG 80 for supply voltages up to and including 440 V (see motor nameplate) must be protected against voltage peaks higher than 650 V between the supply terminals.

### Caution

### Grundfos motors

All three-phase Grundfos motors from frame size 90 and up can be connected to a frequency converter.

The connection of a frequency converter will often have the effect that the motor insulation system is loaded more and that the motor will be more noisy than during normal operation. In addition, large motors are more at risk of being loaded with bearing currents caused by the frequency converter.

In the case of frequency converter operation, consider the following:

- In 2-pole motors from 45 kW, 4-pole motors from 30 kW and 6-pole motors from 22 kW, one of the motor bearings must be electrically insulated to prevent damaging currents from passing through the motor bearings.
- In the case of noise-critical applications, the motor noise can be reduced by fitting an output filter between the motor and the frequency converter. In particularly noise-critical applications, we recommend that you fit a sinusoidal filter.
- The length of the cable between motor and frequency converter affects the motor load. Therefore, check that the cable length meets the specifications laid down by the frequency converter supplier. For supply voltages between 500 and 690 V, either fit a sinusoidal filter to reduce voltage peaks or use a motor with reinforced insulation.
- For supply voltages of 690 V, use a motor with reinforced insulation and fit a sinusoidal filter.

Grundfos MG motors do not have reinforced insulation. When it comes to reinforced insulation, other motor suppliers are able to supply such motors as FPV variants.

### Note

### 6.1.1 Other motor makes than Grundfos

Contact Grundfos or the motor manufacturer.

## 7. Startup

### 7.1 Flushing the pipe system

The pump is not designed to pump liquids containing solid particles such as pipe debris and welding slag. Before starting up the pump, the pipe system must be thoroughly cleaned, flushed and filled with clean water.

**Caution**

The warranty does not cover any damage caused by flushing the pipe system by means of the pump.

### 7.2 Priming

Do not start the pump until it has been filled with liquid and vented. To ensure correct venting, the vent screw must point upwards.

**Caution**

#### Closed systems or open systems where the liquid level is above the pump inlet:

1. Close the isolating valve on the outlet side and loosen the vent screw in the motor stool. See fig. 15.

#### Warning

Pay attention to the direction of the vent hole and make sure that the escaping liquid does not cause injury to persons or damage to the motor or other components. In hot-liquid installations, pay special attention to the risk of injury caused by scalding hot liquid.

In cold-liquid installations, pay special attention to the risk of injury caused by the cold liquid.



2. Slowly open the isolating valve on the inlet side until a steady stream of liquid runs out of the vent hole.
3. Tighten the vent screw and completely open the isolating valve(s).

#### Open systems where the liquid level is below the pump inlet

The inlet pipe and the pump must be filled with liquid and vented before the pump is started.

1. Close the isolating valve on the outlet side and open the isolating valve on the inlet side.
2. Loosen the vent screw. See fig. 15.
3. Remove the plug from one of the pump flanges, depending on the pump location.
4. Pour liquid through the priming hole until the inlet pipe and the pump are filled with liquid.
5. Replace the plug and tighten securely.
6. Tighten the vent screw.

The inlet pipe can to some extent be filled with liquid and vented before it is connected to the pump. A priming device can also be installed before the pump.

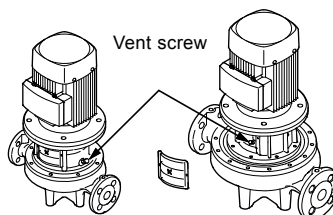


Fig. 15 Position of vent screw

### 7.3 Checking the direction of rotation

Do not start the pump to check the direction of rotation until it has been filled with liquid.

**Note**

Do not check the direction of rotation with the motor alone, as an adjustment of the shaft position is required when the coupling has been removed.

The correct direction of rotation is shown by arrows on the motor fan cover or on the pump housing.

TM03 8126 0507

7.4 Starting

- 1. Before starting the pump, completely open the isolating valve on the inlet side of the pump and leave the isolating valve on the outlet side almost closed.
- 2. Start the pump.
- 3. Vent the pump during starting by loosening the vent screw in the motor stool until a steady stream of liquid runs out of the vent hole. See fig. 15.

Warning

Pay attention to the direction of the vent hole and make sure that the escaping liquid does not cause injury to persons or damage to the motor or other components.



In hot-liquid installations, pay special attention to the risk of injury caused by scalding hot liquid.

In cold-liquid installations, pay special attention to the risk of injury caused by the cold liquid.

- 4. When the pipe system has been filled with liquid, slowly open the isolating valve on the outlet side until it is completely open.

7.5 Shaft seal run-in

The seal faces are lubricated by the pumped liquid, meaning that there may be a certain amount of leakage from the shaft seal.

When the pump is started up for the first time, or when a new shaft seal is installed, a certain run-in period is required before the leakage is reduced to an acceptable level. The time required for this depends on the operating conditions, i.e. every time the operating conditions change, a new run-in period will be started.

Under normal conditions, the leaking liquid will evaporate. As a result, no leakage will be detected. However, liquids such as kerosene will not evaporate. The leakage may therefore be seen as a shaft seal failure.

7.6 Frequency of starts and stops

| Frame size | Maximum number of starts per hour |     |     |
|------------|-----------------------------------|-----|-----|
|            | Number of poles                   |     |     |
|            | 2                                 | 4   | 6   |
| 56-71      | 100                               | 250 | 350 |
| 80-100     | 60                                | 140 | 160 |
| 112-132    | 30                                | 60  | 80  |
| 160-180    | 15                                | 30  | 50  |
| 200-225    | 8                                 | 15  | 30  |
| 250-315    | 4                                 | 8   | 12  |

- On twin-head pumps, the duty and standby pumps must be alternated on a regular basis, i.e. once a week, to ensure an even distribution of the operating hours on both pumps. Pump change can be effected either manually or automatically by installing a suitable pump controller.
- If twin-head pumps are used for pumping domestic hot water, the duty and standby pumps must be alternated on a regular basis, i.e. once a day, to avoid blocking of the standby pump due to deposits (calcareous deposits, etc.). We recommend automatic pump change.

8. Maintenance and service

Warning

Before starting work on the pump, make sure that the power supply has been switched off and that it cannot be accidentally switched on.



Make sure that the escaping liquid does not cause injury to persons or damage to the motor or other components.

In hot-liquid installations, pay special attention to the risk of injury caused by scalding hot liquid.

In cold-liquid installations, pay special attention to the risk of injury caused by the cold liquid.

8.1 Pump

The pump is maintenance-free.

If the pump is to be drained for a long period of inactivity, inject a few drops of silicone oil on the shaft between the motor stool and the coupling. This will prevent the shaft seal faces from sticking.

## 8.2 Motor

Check the motor at regular intervals. It is important to keep the motor clean in order to ensure adequate ventilation. If the pump is installed in a dusty environment, both pump and motor must be cleaned and checked regularly.

### Lubrication

The bearings of motors up to 11 kW are greased for life and require no lubrication.

The bearings of motors of 11 kW and up must be greased in accordance with the indications on the motor nameplate.

Lubricate the motor with a lithium-based, high-temperature grease.

- The technical specification of the grease must correspond to DIN 51825, K3N, or better.
- The viscosity of the basic oil must be higher than 50 cSt (mm<sup>2</sup>/s) at 40 °C (104 °F) and 8 cSt (mm<sup>2</sup>/s) at 100 °C (212 °F).
- The grease filling rate must be 30-40 %.

## 8.3 Service



### Warning

If the pump has been used for a liquid which is injurious to health or toxic, the pump will be classified as contaminated.

If Grundfos is requested to service the pump, Grundfos must be contacted with details about the pumped liquid, etc. before the pump is returned for service. Otherwise Grundfos can refuse to accept the pump for service.

Possible costs of returning the pump are paid by the customer.

## 8.4 Adjusting the shaft

If the motor has been removed during installation or for repair of the pump, the pump shaft must be adjusted after the motor has been replaced.

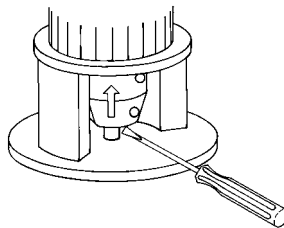
### 8.4.1 Pumps with two-part coupling

#### Pumps Series 100 and 200

Make sure that the shaft pin is fitted in the pump shaft.

Adjust the pump shaft as follows:

1. Remove the coupling guards using a screwdriver.
2. Fit the hexagon socket head screws in the coupling and leave loose.
3. Raise the coupling and the pump shaft as far as possible (towards the motor) with a screwdriver or a similar tool so that the pump and motor shafts touch each other. See fig. 16.

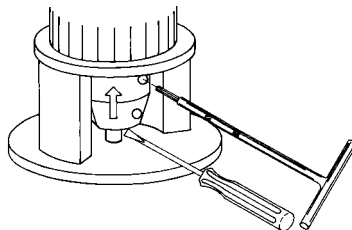


**Fig. 16** Raising the coupling and the pump shaft

4. Tighten the hexagon socket head screws in the coupling to 5 Nm (0.5 kpm).
5. Check that the gaps either side of the coupling halves are equal.
6. Tighten the screws two and two (one side at a time) to the torque stated below. See fig. 17.

| Hexagon socket head screw | Torque          |
|---------------------------|-----------------|
| M6 x 20                   | 13 Nm (1.3 kpm) |
| M8 x 25                   | 31 Nm (3.1 kpm) |

7. Fit the coupling guards.



**Fig. 17** Tightening the screws

### 8.4.2 Pumps with integral shaft/coupling

For pumps with integral shaft/coupling, we recommend that you do not remove the motor. If the motor has been removed, it is necessary to remove the motor stool in order to refit the motor correctly. Otherwise the shaft seal may be damaged.

TM00 6415 3695

TM00 6416 3695

8.5 Blanking flanges

For twin-head pumps, a blanking flange with a pump housing gasket is available. See fig. 18.

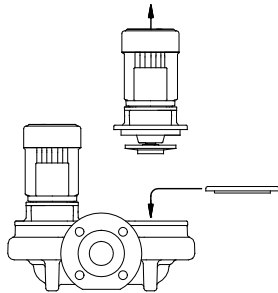


Fig. 18 Fitting the blanking flange

If one pump requires service, the blanking flange is fitted to allow the other pump to continue operating.

9. Technical data

9.1 Ambient temperature

Maximum 55 °C (131 °F).

9.2 Liquid temperature

-40 - +150 °C (-40 - +302 °F).

The maximum liquid temperature depends on the mechanical shaft seal type and the pump type.

Depending on the cast-iron version and the pump application, the maximum liquid temperature may be limited by local regulations and laws.

The maximum liquid temperature is marked on the pump nameplate.

**Note**

If the pump is operating with liquids at high temperatures, the life of the shaft seal may be reduced. It may be necessary to replace the shaft seal more often.

9.3 Operating pressure/test pressure

The pressure test has been made with water containing anti-corrosive additives at a temperature of 20 °C (68 °F).

| Pressure stage | Operating pressure |       | Test pressure |       |
|----------------|--------------------|-------|---------------|-------|
|                | [bar]              | [MPa] | [bar]         | [MPa] |
| PN 6           | 6                  | 0.6   | 10            | 1.0   |
| PN 6 / PN 10   | 10                 | 1.0   | 15            | 1.5   |
| PN 16          | 16                 | 1.6   | 24            | 2.4   |
| PN 25          | 25                 | 2.5   | 38            | 3.8   |

9.4 Inlet pressure

To ensure optimum and quiet pump operation, the inlet pressure (system pressure) must be adjusted correctly. See the table on page 31.

For the calculation of specific inlet pressures, contact the local Grundfos company or see the data booklet for TP, TPD, TPE, TPED, TPE2, TPE2 D, TPE3, TPE3 D, if at hand.

9.5 Enclosure class

Closed drain hole in motor: IP55.

Open the drain hole in motor: IP44. See fig. 9.

9.6 Electrical data

See motor nameplate.

9.7 Sound pressure level

Pumps with single-phase motors

The sound pressure level of the pump is lower than 70 dB(A).

Pumps with three-phase motors

See the table on page 42.

9.8 Environment

Non-aggressive and non-explosive atmosphere.

Relative air humidity: Maximum 95 %.

TM00 6360 3495

## 10. Fault finding the product

### Warning

Before removing the terminal box cover and before removal/dismantling the pump, make sure that the power supply has been switched off and that it cannot be accidentally switched on.



Make sure that the escaping liquid does not cause injury to persons or damage to the motor or other components.

In hot-liquid installations, pay special attention to the risk of injury caused by scalding hot liquid.

In cold-liquid installations, pay special attention to the risk of injury caused by the cold liquid.

| Fault                                                                          | Cause                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Motor does not run when started.                                            | a) Power supply failure.<br>b) Fuses blown.<br>c) Motor protection device has tripped.<br>d) Main contacts in motor protection device are not making contact or the coil is faulty.<br>e) Control circuit fuses are defective.<br>f) Motor is defective. |
| 2. Motor protection device trips immediately when power supply is switched on. | a) Power supply failure.<br>b) Contacts in motor protection device are faulty.<br>c) Cable connection is loose or faulty.<br>d) Motor winding is defective.<br>e) Pump is mechanically blocked.<br>f) Overload setting too low.                          |
| 3. Motor protection device trips occasionally.                                 | a) Overload setting too low.<br>b) Supply voltage periodically too low or too high.<br>c) Differential pressure across pump too low.                                                                                                                     |
| 4. Motor protection device has not tripped but the pump does not run.          | a) Check the power supply.<br>b) Check fuses.<br>c) Check the main contacts in the motor protection device and coil.<br>d) Check the control circuit.                                                                                                    |
| 5. Pump capacity not constant.                                                 | a) Pump inlet pressure is too low.<br>b) Inlet pipe/pump partly blocked by impurities.<br>c) Pump draws in air.                                                                                                                                          |
| 6. Pump runs but gives no water.                                               | a) Inlet pipe/pump blocked by impurities.<br>b) Foot or non-return valve blocked in closed position.<br>c) Leakage in the inlet pipe.<br>d) Air in inlet pipe or pump.<br>e) Motor rotates in the wrong direction.                                       |
| 7. Pump runs backwards when switched off.*                                     | a) Leakage in the inlet pipe.<br>b) Foot or non-return valve defective.<br>c) Foot or non-return valve blocked in open or partly open position.                                                                                                          |
| 8. Leakage in the shaft seal.                                                  | a) Pump shaft position is incorrect.<br>b) Shaft seal is defective.                                                                                                                                                                                      |

| Fault                                                                                  | Cause                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9. Noise.                                                                              | <div>a) Pump is cavitating.</div> <div>b) Pump does not rotate freely (frictional resistance) because of incorrect pump shaft position.</div> <div>c) Frequency converter operation:<br/>See section <a href="#">6.1 Frequency converter operation</a>.</div> <div>d) Resonance in the installation.</div> <div>e) Foreign bodies in the pump.</div>                                    |
| 10. Pump runs constantly (applies only to pumps with automatic start/stop).            | <div>a) Stop pressure is too high in relation to the required quantity of water.</div> <div>b) The water consumption is larger than anticipated.</div> <div>c) Leakage in the outlet pipe.</div> <div>d) The direction of rotation of the pump is incorrect.</div> <div>e) Pipes, valves or strainer blocked by impurities.</div> <div>f) Pump controller, if used, is defective.</div> |
| 11. Period of operation is too long (applies only to pumps with automatic start/stop). | <div>a) Stop pressure is too high in relation to the required quantity of water.</div> <div>b) Pipes, valves or strainer blocked by impurities.</div> <div>c) Pump partly blocked or seized up.</div> <div>d) The water consumption is larger than anticipated.</div> <div>e) Leakage in the outlet pipe.</div>                                                                         |

\* In twin-head pump installations, the standby pump will often rotate slowly.

11. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.



## 中文 (CN) 安装和使用说明书

## 中文版本

## 目录

|               | 页  |
|---------------|----|
| 1. 本文献中所用符号   | 17 |
| 2. 概述         | 17 |
| 3. 运输和吊装      | 18 |
| 3.1 运输        | 18 |
| 3.2 吊装        | 18 |
| 4. 应用         | 19 |
| 4.1 泵送液体      | 19 |
| 5. 安装         | 19 |
| 5.1 管路连接      | 21 |
| 5.2 降噪和减振     | 21 |
| 5.3 基础        | 22 |
| 5.4 接线盒位置     | 23 |
| 5.5 底座        | 23 |
| 5.6 隔热        | 23 |
| 5.7 霜冻防护      | 23 |
| 6. 电气连接       | 24 |
| 6.1 变频器操作     | 24 |
| 7. 启动         | 25 |
| 7.1 冲洗管道系统    | 25 |
| 7.2 启动注水      | 25 |
| 7.3 检查旋转的方向   | 25 |
| 7.4 启动        | 26 |
| 7.5 轴封磨合      | 26 |
| 7.6 启动和停机频率   | 26 |
| 8. 保养和服务      | 26 |
| 8.1 泵         | 26 |
| 8.2 电机        | 27 |
| 8.3 维护        | 27 |
| 8.4 轴调节       | 27 |
| 8.5 盲板法兰      | 28 |
| 9. 技术数据       | 28 |
| 9.1 环境温度      | 28 |
| 9.2 液体温度      | 28 |
| 9.3 操作压力和试验压力 | 28 |
| 9.4 入口压力      | 28 |
| 9.5 防护等级      | 28 |
| 9.6 电气数据      | 28 |
| 9.7 声压级       | 28 |
| 9.8 环境        | 28 |
| 10. 故障排除      | 29 |
| 11. 回收处理      | 30 |

**警告**

装机前，先仔细阅读本安装操作手册。安装和运行必须遵守当地规章制度并符合公认的良好操作习惯。

**1. 本文献中所用符号****警告**

不执行这些安全须知可能会引起人身伤害。

**小心**

不遵守这些指导可能会造成设备故障或设备损坏。

**注意**

遵守注意事项或使用说明可以简化作业并可保证操作安全。

**2. 概述**

本手册适用于配备有格兰富电机的TP和TPD型水泵。如果水泵安装了其它类型的电机，务必要注意其电机资料可能会与本说明手册不同。

## 3. 运输和吊装

### 3.1 运输

泵出厂时装在木底箱子中，这是专为使用叉车或类似车辆进行运送而设计。

### 3.2 吊装

#### 警告



大型水泵电机配备的吊眼可用于起吊泵头（电机、电机托架和叶轮）。吊眼不得用于起吊整个水泵。

TPD：水泵外壳中央的线索不能用来起吊，这是因为线索位于水泵的重心之下。

没有吊环的泵必须使用尼龙带进行吊装。见图1和2。

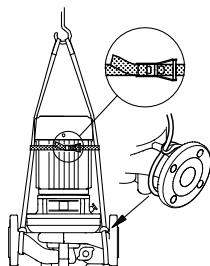


图 1 TP

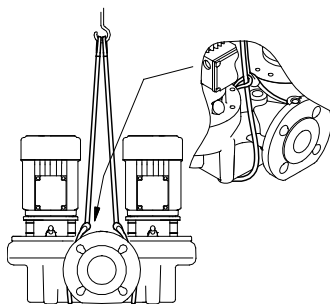


图 2 TPD

有吊环的泵必须使用尼龙带和锁紧卡环进行吊装。见图3和4。

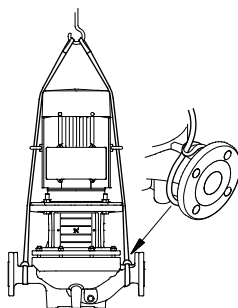


图 3 TP

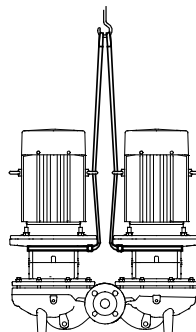


图 4 TPD

TM02 7009 2303

TM02 7007 2303

TM02 7010 2303

TM02 7008 2303

## 4. 应用

该泵用于在住宅、机构和工业应用中循环热水或冷水，用于以下系统：

- 加热系统
- 区域供暖站
- 公寓楼的集中供暖系统
- 空调系统
- 冷却系统。

此外，本系列水泵还适用于液体输送和给水系统，例如：

- 冲洗系统
- 家用热水系统
- 一般的工业系统。

为了达到最佳工作状态，系统的工作范围必须处于水泵的性能范围之内。

### 4.1 泵送液体

泵送的液体应该具有易流动、粘度小、清洁、无腐蚀性、无爆炸性，不得含有对水泵造成机械或者化学损伤的固体颗粒物质或者纤维。

举例：

- 集中供热系统(泵送水质应符合公认的供热系统水质标准要求)
- 冷却液
- 家用热水
- 工业用液体
- 软化水。

如果泵送液体的密度和/或运动粘度高于水的密度和运动粘度，将会导致：

- 压力大幅度下降
- 水力性能降低
- 能源消耗增加。

在此情况下，泵应配备更大的电机。如有疑问，请联系格兰富公司。

作为标准安装的EPDM O型圈主要适用于水。

如果水中含有矿物质、合成油或者化学物质，或者泵送的液体是水以外的其它液体，应相应地选择O型圈。

## 5. 安装



### 警告

当泵的工作液体为高温或低温时，请注意避免人员意外接触到高温或低温表面。

水泵必须放置于干燥、通风且无霜冻的场所。

用水泵法兰(PN 6/10)中的椭圆螺栓孔安装水泵时，要使用如图5所示的垫圈。

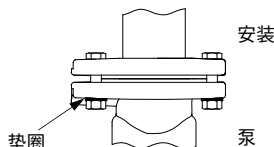


图5 在椭圆螺栓孔上使用垫圈

泵壳体上的箭头表示液体通过泵的流通方向。

电机功率最大在11千瓦以下的水泵可以安装在水平或者垂直的管道系统中。

电机功率等于或者大于11千瓦的水泵只能安装在水平的管道系统中，而且电机应该垂直放置。

然而，一些11 kW及以上的TP、TPE泵可直接悬挂在管道(水平或垂直)中。见第43页上的表*TP, TPE pumps from 11 kW and up suspended in the pipes*。

在泵直接悬挂在管道中的情况下，泵可以支持泵两端长度为L的管道( $L < 3 \times DN$ )。见图6。在泵直接悬挂在管道中的情况下，泵必须用绳索或类似的装置吊起并保持在正确的位置上，直到两个泵法兰被完全紧固到管法兰。

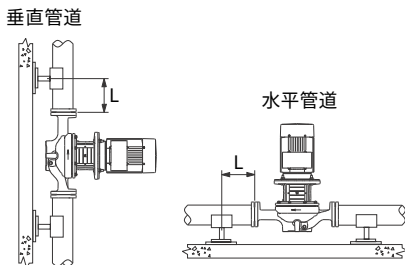


图6 直接悬挂在管道中的泵

TM01 0683 1997

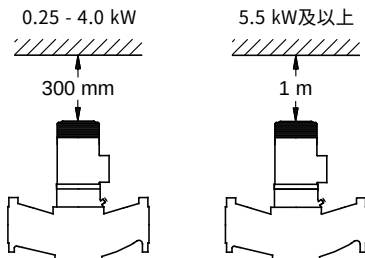
TM06 3518 0615

**小心** 电机不得低于水平面。

为了检测和拆卸电机/水泵头，对电机上方的空隙规定如下：

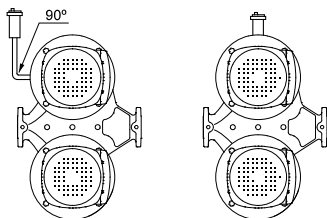
- 功率为4.0千瓦及以下的电机为300 mm。
- 功率为5.5千瓦及以上的电机为1米。

见图7。



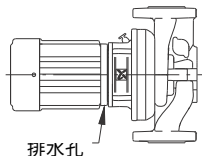
**图 7** 电机上方的空隙

安装在水平管里的双头水泵必须在水泵外壳的上部分安装一个自动排气阀。见图8。自动排气阀不随泵一起供应。



**图 8** 自动排气阀

如果液体的温度降低到环境温度以下，水泵停机后在电机里可能出现凝水现象。在这种情况下须确保打开电机法兰的排水孔，使其朝下。见图9。



**图 9** 电机法兰的排水孔

若使用双头水泵泵送温度低于0 °C / 32 °F的液体，冷凝水可能会结冰并导致联轴器无法运转。安装加热元件可防止冻结问题。如有可能(电机功率低于11千瓦的水泵)，水泵安装时，电机轴应处于水平位置。见图8。

**小心** 务必遵循章节9. [技术数据](#) 技术数据的说明要求。

TM00 9831 4715

TM00 3733 2802

TM03 8127 0507

5.1 管路连接

在水泵两侧安装隔离阀，如果水泵需要清理或维修，则可以避免进行系统排水。

如果水泵两侧的管道系统能被充分支撑起来，水泵可以进行管线安装。TP 25-50, 25-80, 25-90, 32-50, 32-80, 32-90, 40-50, 40-80和40-90专门用于管线安装。

安装管路时，务必确保泵壳不会受到管道的牵拉。

吸水管和排水管的尺寸必须恰当，同时需要考虑到泵的入口压力。

不得将水泵安装在系统的最低点以避免产生污物沉积。

安装管道时应注意避免气穴形成，特别是在泵的吸入侧。见图10。

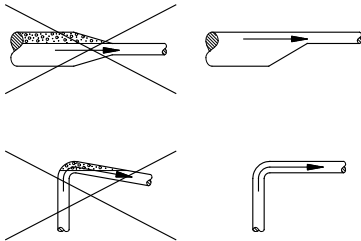


图 10 在水泵的吸水侧安装正确的管道。

小心

不要在出口阀关闭时运转水泵，因为这可能导致温度升高/泵内液体汽化，从而损坏水泵。

如果水泵存在任何在出口阀关闭情况下运行的可能，应该在出口阀前的排出管路上连接一个旁通 / 排水管以确保最小流量得以流经水泵。比如，排水管可以与一个水箱连接。始终确保水泵的最低流量为其最高效率工作时流量的10 %。

最大效率工作时流速和扬程请详见水泵铭牌。

5.2 降噪和减振

为了达到最佳工作状态，将噪音和振动降至最低，可考虑为水泵安装减振器。一般来说，对于电机功率不小于11千瓦的水泵需要考虑安装减振器，而对于功率在90千瓦及以上的水泵以及下表中列出的水泵则必须安装减振器。

| 泵型           | P2 [kW] | 频率 [赫兹] |
|--------------|---------|---------|
| TP 200-280/4 | 37      | 60      |
| TP 200-290/4 | 37      | 50      |
| TP 200-320/4 | 45      | 60      |
| TP 200-360/4 | 55      | 60      |
| TP 200-390/4 | 75      | 60      |

较小功率的电机也会发生较大的噪音和振动。

噪音和振动是由于电机和水泵运转，以及水管和管道配件中的水流造成的。对环境造成的影响取决于安装是否正确以及其余系统的工作状态。

降噪减振最好的方法是使用混凝土基础、减振器和伸缩接头。

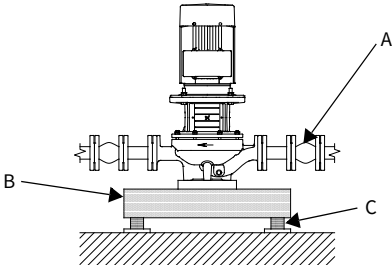


图 11 TP水泵基座

| 序号 | 描述    |
|----|-------|
| A  | 伸缩接头  |
| B  | 混凝土基座 |
| C  | 减振器   |

在高液体速度(> 5米/秒)的情况下, 建议使用适合管道的较大伸缩接头。

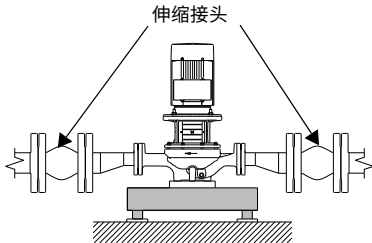


图 12 安装较大的伸缩接头的TP水泵

5.3 基础

格兰富建议您将水泵安装在一个混凝土底座上, 底座的重量应该足以对整个泵提供永久性的牢固支撑。基础必须能够吸收振动, 正常的应力或冲击。根据经验, 混凝土底座的重量必须为泵总重的1.5倍。将泵放置在基础上, 将其固定。见图11。

5.3.1 建议对TP、TPD 300系列的水泵使用混凝土底座

对于重150千克或以上的TP 300系列泵, 我们建议您将泵安装在下表所列尺寸的混凝土底座上。对于TPD 300系列的水泵中重量等于或大于300千克的水泵, 该建议同样适用。

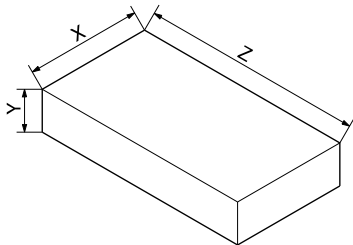


图 13 TP、TPD 300系列泵的底座

混凝土基础尺寸规格

| 水泵重量<br>[kg] | Y<br>(高度)<br>[mm] | Z<br>(长度)<br>[mm] | X<br>(宽度)<br>[mm] |
|--------------|-------------------|-------------------|-------------------|
| 150          | 280               | 565               | 565               |
| 200          | 310               | 620               | 620               |
| 250          | 330               | 670               | 670               |
| 300          | 360               | 710               | 710               |
| 350          | 375               | 750               | 750               |
| 400          | 390               | 780               | 780               |
| 450          | 410               | 810               | 810               |
| 500          | 420               | 840               | 840               |
| 550          | 440               | 870               | 870               |
| 600          | 450               | 900               | 900               |
| 650          | 460               | 920               | 920               |
| 700          | 470               | 940               | 940               |
| 750          | 480               | 970               | 970               |
| 800          | 490               | 990               | 990               |
| 850          | 500               | 1010              | 1010              |
| 900          | ≤ DN<br>200       | 510               | 1030              |
| 950          |                   | 520               | 1050              |
| 1000         |                   | 530               | 1060              |
| 1050         |                   | 540               | 1080              |
| 1100         |                   | 550               | 1100              |
| 1150         |                   | 560               | 1100              |
| 1200         |                   | 560               | 1130              |
| 1250         |                   | 570               | 1150              |
| 1300         |                   | 580               | 1160              |
| 1350         |                   | 590               | 1180              |
| 1400         |                   | 600               | 1190              |
| 1450         |                   | 600               | 1200              |
| 1500         |                   | 610               | 1220              |
| 1550         |                   | 620               | 1230              |
| 1600         |                   | 620               | 1250              |
| 1650         |                   | 630               | 1250              |
| 1700         |                   | 635               | 1270              |

混凝土基础尺寸规格

| 水泵重量<br>[kg] | Y<br>(高度)<br>[mm]              | Z<br>(长度)<br>[mm] | X<br>(宽度)<br>[mm] |
|--------------|--------------------------------|-------------------|-------------------|
| 800          | 450                            | 1400              | 800               |
| 1000         | 450                            | 1400              | 1000              |
| 1200         | 450                            | 1400              | 1200              |
| 1400         | 500                            | 1600              | 1200              |
| 1600         | 500                            | 1600              | 1350              |
| 1800         | 500                            | 1600              | 1500              |
| 2000         | 550                            | 1600              | 1600              |
| 2200         | 550                            | 1700              | 1700              |
| 2400         | DN 300 /<br>DN 350 /<br>DN 400 | 550               | 1800              |
| 2600         |                                | 600               | 1800              |
| 3000         |                                | 600               | 2000              |
| 3400         |                                | 680               | 2000              |
| 3800         |                                | 760               | 2000              |
| 4200         |                                | 840               | 2000              |
| 4600         |                                | 920               | 2000              |
| 5000         |                                | 1000              | 2000              |
| 5400         |                                | 1080              | 2000              |

## 5.4 接线盒位置



### 警告

在泵启动工作前，应确保电源断开，且不会被意外接通。

端子盒可以转到四个位置的任何一个位置，四个位置以90°分布。

按照下列操作步骤改变端子盒的位置：

1. 必要时，用螺丝起子拆除联轴器护板。不得拆除联轴器。
2. 拆下用来将电机固定在水泵上的螺丝。
3. 将电机转到需要的位置。
4. 重新安装并旋紧螺丝。
5. 重新安装联轴器护板。

## 5.5 底座

单头水泵(除TP 25-50, 25-80, 25-90, 32-50, 32-80, 32-90, 40-50, 40-80, 40-90 外)在水泵外壳的底部有两个螺丝孔，可以用来将格兰富底座安装到水泵上。底座为选用附件。

双头水泵在水泵外壳的底部有四个螺丝孔。对一些双头水泵，可以用两个半底座。

底座的尺寸见第46页。

## 5.6 隔热

小心

勿对电机托架进行密封，这样会阻碍蒸汽从轴密封排出，从而导致腐蚀。在电机托架盖上密封之后也会使检查和维修变得困难。

在对泵进行隔离时，按照图14中指示的进行操作。

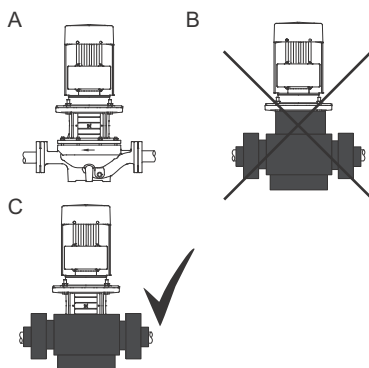


图 14 TP泵隔离

| 序号 | 描述      |
|----|---------|
| A  | 无绝缘     |
| B  | 错误的隔离密封 |
| C  | 正确隔离密封  |

## 5.7 霜冻防护

如果在严寒季节不需要使用水泵时，应该排空水泵以防损坏。

TM05 2328 4911

## 6. 电气连接

遵照当地法规进行电气连接。



### 警告

在拆去接线盒盖以及任何对泵的拆卸/拆装工作开展之前，确保电源已经断开。

将水泵连接到外部电源开关，其电极间的接触间隙最小为3 mm。

水泵的铭牌上标有适用的运行电压和频率。确保电动机适用于其供应电源。

单相标准电机配有一只热控开关，因此不需要额外的电机保护。

三相电机必须连接到电机-保护断路器上。

功率为3千瓦及以上的电机配有热敏电阻(PTC)。热敏电阻设计符合DIN 44082设计标准。

按照接线盒盖内的接线图所示进行电气连接。

双头水泵的电机应单独连接。

### 小心

在加满液体和排气之前不要启动水泵。

## 6.1 变频器操作

### 小心

电源供应电压为440伏及440伏以下的西门子，MG 71和MG 80型号的电机(见电机铭牌)必须加以保护以避免出现两电源端子之间的电压的峰值高于650伏的情况。

### 格兰富电机

所有机架尺寸大于90的三相电机都可以与一个变频器相连。

水泵连接变频器工作时会增加电动机绝缘的负载以及电动机运行噪音。此外，大功率电机还会因为轴电流而增加负载。

使用变频器时，请注意以下问题：

- 在45 kW 2极电机、30 kW 4极电机和22 kW 6极电机中，一个电机轴承必须绝缘，以防止有害电流穿过电机轴承。
- 对于需要保持较小噪音的应用，可以在电机与变频器之间串联一个输出滤波器来减小噪音。对于特别需要尽可能减小噪音的应用，我们建议您安装一个正弦波滤波器。
- 电机与变频器之间的电缆长度会影响电机负载。因此，应确认电缆长度符合变频器供应商制定的规格要求。如果电源电压在500 V到690 V之间，则可以安装正弦波滤波器降低峰值电压，或使用带有加强绝缘的电机。
- 如果电源电压为690 V，则使用带有加强绝缘的电机，并安装一个正弦波滤波器。

### 注意

格兰富MG电机不具有加强绝缘层。对于加强绝缘层，其他电机供应商能够提供类似于FPV型号的电机。

### 6.1.1 非格兰富制造的电机。

联系格兰富公司或电机制造商。



## 7. 启动

### 7.1 冲洗管道系统

**小心**

水泵不适合用于泵送含有管道碎片和焊渣等固体颗粒的液体。启动泵之前，管道系统必须彻底清洁和冲洗，并注满干净的水。如因冲洗管道系统导致水泵受损，质保对此无效。

### 7.2 启动注水

**小心**

在加满液体和排气之前不要启动水泵。排气螺丝应向上，以保证正确的排气。

#### 液面水平高于水泵进口的闭式系统或者开式系统：

1. 关闭排放截止阀，并拧松电机托架内的排气螺丝。见图15。

#### 警告

注意排气口的方向，务必确保流出的水不会对人员造成损伤或者对电机或其它元件造成损坏。

在高温液体安装中，要特别注意防止灼热的液体造成人身伤害等风险。

在低温液体安装中，要特别注意由低温液体引起的人身伤害风险。

2. 慢慢打开进水管里的截止阀，直到稳定的水流通过排气口为止。
3. 旋紧排气螺丝，完全打开截止阀。

#### 液面水平在水泵进口以下的开式系统：

水泵启动前，进水管和水泵必须充满液体并充分排气。

1. 关闭出口阀，打开进水管里的截止阀。
2. 拧松排气螺丝。见图15。
3. 拔出水泵任一法兰的塞子，根据水泵的位置来确定。
4. 通过注水孔注入液体直到吸水管和水泵均充满液体。
5. 重新插上塞子并旋紧。
6. 旋紧排气螺丝。

进水管在连接到水泵之前可以一定程度地充满液体并排气。在水泵前可以安装一个注水器。

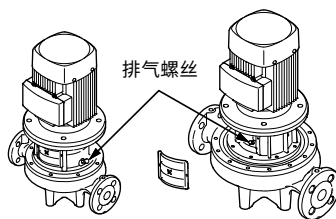


图 15 排气螺丝位置

### 7.3 检查旋转的方向

在泵充满液体之前，不得启动泵检查旋转的方向。

**注意**

不应只通过电机检查旋转方向，因为取出联轴器之后，轴的位置需要进行调整。

正确的旋转方向必须与电机风扇盖或者水泵外壳上的箭头方向一致。

TM03 8126 0507

7.4 启动

- 1. 启动水泵前，进水侧的截止阀门完全打开，而出水口截止阀门几乎完全关闭。
- 2. 启动水泵。
- 3. 启动时，拧松电机托架内的排气螺丝使水泵排气，直到通气孔的液体流量稳定。见图15。

警告



注意排气口的方向，务必确保流出的水不会对人员造成损伤或者对电机或其它元件造成损坏。

在高温液体安装中，要特别注意防止灼热的液体造成人身伤害等风险。

在低温液体安装中，要特别注意由低温液体引起的人身伤害风险。

- 4. 当管道系统充满液体之后，慢慢打开出水口截止阀门直到其完全打开。

7.5 轴封磨合

泵送液体对轴封的密封面进行润滑，也就意味着密封处可能会有一定量的渗漏。

首次启动水泵，或者安装新的轴封时，需经过一定的磨合周期后，渗漏才会降至合理水平。所需时间取决于运行状况，即每当运行状况改变时，就需要开始新的磨合期。

在通常条件下，渗漏液体会汽化。因此，不会检查到渗漏。

但是有些液体，例如煤油，不会汽化。因此，渗漏可能会被认为轴封发生故障。

7.6 启动和停机频率

| 每小时最大启动次数 |     |     |     |
|-----------|-----|-----|-----|
| 机架尺寸      | 极数  |     |     |
|           | 2   | 4   | 6   |
| 56-71     | 100 | 250 | 350 |
| 80-100    | 60  | 140 | 160 |
| 112-132   | 30  | 60  | 80  |
| 160-180   | 15  | 30  | 50  |
| 200-225   | 8   | 15  | 30  |
| 250-315   | 4   | 8   | 12  |

- 在双头水泵上，正在运行的水泵和处于待机状态的水泵应该定期轮流运行，如一周轮换一次，以保证两个水泵的运行时间均等。交换使用水泵可以采用手工控制来完成，也可以通过安装一个适当的水泵控制器自动完成。
- 若双头水泵用于泵送生活用热水，正在运行的水泵和处于待机状态的水泵应该定期轮流运行，如一天交换一次，以避免待机水泵因为沉积物(如钙质沉积物等)积累而堵塞。我们建议进行自动泵更换。

8. 保养和服务

警告

在泵启动工作前，应确保电源断开，且不会被意外接通。



确保溢出的水不会对人员造成伤害或损坏电机或其它部件。

在高温液体安装中，要特别注意防止灼热的液体造成人身伤害等风险。

在低温液体安装中，要特别注意由低温液体引起的人身伤害风险。

8.1 泵

水泵是免维护的。

如果水泵因为长时间未用而排空，在电机支架和联轴器之间的轴上注入几滴硅油。这将有效地防止了轴端密封的粘结。

## 8.2 电机

定期检查电机。为保证通风充足，请务必保持电机清洁。如果泵安装在灰尘较多的环境中，必须定期清洗和检查水泵和电机。

### 润滑

功率为11千瓦以下的电机轴承上添加的润滑脂可以满足电机整个使用寿命内的需求，因此不需要再润滑。

功率为11千瓦及以上的电机的轴承必须根据电机铭牌上的说明加注润滑脂。

使用锂基高温润滑脂润滑电机。

- 润滑油脂的技术规格必须符合DIN 51825的规定，必须为K3N或更佳。
- 在40 °C (104 °F) 下，基础油的粘度必须大于50 cSt (mm<sup>2</sup>/s)，在 100 °C (212 °F) 下必须大于8 cSt (mm<sup>2</sup>/s)。
- 润滑脂的充填速率必须为30 %至40 %。

## 8.3 维护



### 警告

如果泵用于有害健康的液体或有毒的液体，则泵必须归类为受污染水泵。

如需格兰富进行水泵维修，在水泵被送来维修之前，必须告知格兰富公司有关泵送液体等的详细资料。否则，格兰富有权拒绝对该泵进行维修。

泵返厂可能发生的费用由客户承担。

## 8.4 轴调节

如果在安装过程中或者水泵维修时电机被拆卸，重新安装电机后必须调节水泵轴。

### 8.4.1 两半式联轴器水泵

#### 100, 200 系列水泵

确保轴销与水泵的轴相吻合。

按下列步骤调节水泵轴：

1. 用螺丝起子拆下联轴器护板。
2. 将内六角头螺丝放入联轴器中，先不要拧紧。
3. 用螺丝起子或者类似工具尽可能地抬起联轴器和水泵轴(朝电机方向)，使水泵轴和电机轴互相接触。见图16。

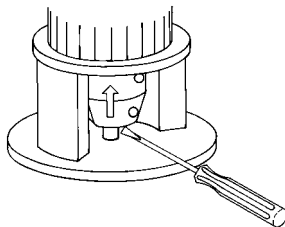


图 16 抬起联轴器和水泵轴

4. 以5 Nm (0.5 kpm) 的转矩将内六角头螺丝安装在联轴器里。
5. 检查半联轴器每侧的间隙相等。
6. 两个两个地旋紧螺丝(每次一边)，转矩如下表：见图17。

| 内六角头螺丝  | 扭矩              |
|---------|-----------------|
| M6 x 20 | 13 Nm (1.3 kpm) |
| M8 x 25 | 31 Nm (3.1 kpm) |

7. 安装联轴器护板。

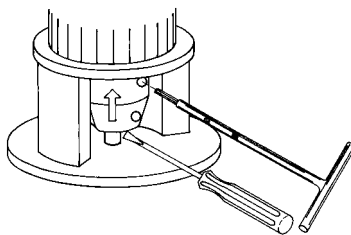


图 17 旋紧螺丝

#### 8.4.2 配有整体轴 / 联轴器的水泵

针对配有整体轴和联轴器的水泵，我们建议不要移除电机。如果电机已经被移出，也必须同时移出电机支架，以便重新正确安装电机。否则轴封将被损坏。

TM00 6415 3695

TM00 6416 3695

8.5 盲板法兰

对于双头水泵，我们可以提供一个带水泵外壳垫圈的盲板法兰。见图18。

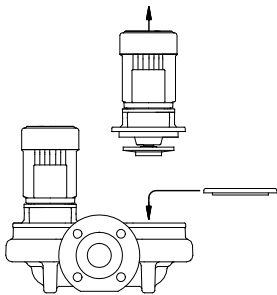


图 18 安装盲板法兰

如果一个水泵需要进行维修，那么可以安装盲板法兰使另一台水泵继续工作。

9. 技术数据

9.1 环境温度

最高温度55 °C (-131 °F)。

9.2 液体温度

-40 °C (-40 °F) 最高至150 °C (-302 °F)。

最高液体温度取决机械轴封的类型和水泵类型。

根据铸铁的类型和水泵用途，液体最大温度还可能受地方法律法规的限制。

最高液体温度在水泵铭牌上已标明。

**注意** 如果水泵用来泵送高温液体，轴封的寿命可能缩短。因此有必要经常更换轴封。

9.3 操作压力和试验压力

压力试验通过含防腐添加剂的水在20 °C (约68 °F) 温度下完成。

| 压力等级         | 运行压力  |       | 测试压力  |       |
|--------------|-------|-------|-------|-------|
|              | [bar] | [MPa] | [bar] | [MPa] |
| PN 6         | 6     | 0.6   | 10    | 1.0   |
| PN 6 / PN 10 | 10    | 1.0   | 15    | 1.5   |
| PN 16        | 16    | 1.6   | 24    | 2.4   |
| PN 25        | 25    | 2.5   | 38    | 3.8   |

9.4 入口压力

为了保证水泵处于最佳工作状态，同时噪音降低到最小的水平，进口压力(系统压力)必须正确调节。见第31页上的表格。

为了计算具体的进口压力，请与当地的格兰富公司联系或者参见TP、TPD、TPE、TPED、TPE2、TPE2 D、TPE3、TPE3 D数据手册。

9.5 防护等级

封闭电机的排水孔：IP55。

打开电机的排水孔：IP44。(排水孔，参见图9。)

9.6 电气数据

参见电机铭牌。

9.7 声压级

单相电机水泵

水泵的声压水平低于70 dB(A)。

三相电机水泵

见第42页上的表格。

9.8 环境

无腐蚀性、无爆炸性空气。

相对空气湿度：最大为95 %。

TM00 6360 3495

## 10. 故障排除

### 警告

拆卸接线盒盖子和拆解水泵前，确保电源已断开，且不会意外接通。

确保溢出的水不会对人员造成伤害或损坏电机或其它部件。

在高温液体安装中，要特别注意防止灼热的液体造成人身伤害等风险。

在低温液体安装中，要特别注意由低温液体引起的人身伤害风险。



| 故障                   | 原因                                                                                                                         |
|----------------------|----------------------------------------------------------------------------------------------------------------------------|
| 1. 电机启动时不能运行。        | a) 电源故障<br>b) 保险丝熔断。<br>c) 电机保护装置跳闸。<br>d) 电机保护装置的主触点未接触或线圈有故障。<br>e) 控制电路保险丝损坏。<br>f) 电机损坏。                               |
| 2. 电源接通时电机保护装置跳闸。    | a) 电源故障<br>b) 电机保护装置触点故障。<br>c) 电缆连接松开或者出现故障。<br>d) 电机绕组损坏。<br>e) 水泵发生机械性堵塞。<br>f) 过载设置太低。                                 |
| 3. 电机保护装置偶尔跳闸。       | a) 过载设置太低。<br>b) 供给电压周期性过低或者过高。<br>c) 通过水泵的压差太低。                                                                           |
| 4. 电机保护装置没有跳闸，但泵不运行。 | a) 接通电源。<br>b) 检查保险丝。<br>c) 检查电机保护装置和线圈的主触点。<br>d) 检查控制电路。                                                                 |
| 5. 泵的容量不恒定。          | a) 水泵进口压力太低。<br>b) 吸水管和水泵被杂物部分堵塞。<br>c) 泵内进气。                                                                              |
| 6. 水泵运行但是不出水。        | a) 吸水管/水泵被杂物部分地堵塞。<br>b) 底阀或者止回阀卡在关闭位置。<br>c) 吸入管泄漏。<br>d) 吸入管或泵进了空气。<br>e) 电机的转动方向错误。                                     |
| 7. 水泵在关机时倒转。*        | a) 吸入管泄漏。<br>b) 底阀或者单向阀门损坏。<br>c) 底阀或者止回阀门卡在全开或者半开的位置。                                                                     |
| 8. 轴封泄漏。             | a) 水泵轴的位置不正确。<br>b) 轴封损坏。                                                                                                  |
| 9. 噪音。               | a) 水泵出现汽穴(汽蚀)现象。<br>b) 水泵不能自由转动(摩擦阻力) 因为水泵轴的位置错误。<br>c) 变频器操作：<br>见 <a href="#">6.1 变频器操作</a> 。<br>d) 装置出现共振。<br>e) 泵中有异物。 |

| 故障                               | 原因                                                                                                                                                               |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10. 水泵持续运行 (仅适用于具有自动启动和停机功能的水泵)。 | <div>a) 相对于所需水量，停机压力太高。</div> <div>b) 水消耗比预期的要大。</div> <div>c) 出水管渗漏。</div> <div>d) 水泵的转动方向错误。</div> <div>e) 管道、阀门或者过滤器被杂物堵塞。</div> <div>f) 水泵控制器(如果采用)损坏。</div> |
| 11. 运行周期过长 (仅适用于具有自动启动和停机功能的水泵)。 | <div>a) 相对于所需水量，停机压力太高。</div> <div>b) 管道、阀门或者过滤器被杂物堵塞。</div> <div>c) 水泵部分堵塞或卡死。</div> <div>d) 水消耗比预期的要大。</div> <div>e) 出水管渗漏。</div>                              |

\* 在双头水泵装置中，处于待机状态的水泵经常慢速转动。

11. 回收处理

必须以环境友好的方式对本产品或产品的部件进行回收处理。

1. 使用公立或私立废品回收服务设施。
2. 如果以上无法做到，与附近的格兰富公司或服务站联系。

内容可有变动。

## Appendix

**GB:** Inlet pressure stated in bar relative pressure (pressure gauge value measured on the suction side of the pump)

**CN:** 入口压力为以巴为单位的相对压力（压力计在水泵进水侧测量到的数值）

**50 Hz, 2-pole**

| Pump type<br>(50 Hz) | p [bar] |       |       |        |        |        |        |
|----------------------|---------|-------|-------|--------|--------|--------|--------|
|                      | 20 °C   | 60 °C | 90 °C | 110 °C | 120 °C | 140 °C | 150 °C |
| TP 25-50R/2          | 0.1     | 0.1   | 0.2   | 0.5    | -      | -      | -      |
| TP 25-80R/2          | 0.1     | 0.1   | 0.1   | 0.3    | -      | -      | -      |
| TP 25-90R/2          | 0.1     | 0.1   | 0.2   | 0.5    | -      | -      | -      |
| TP 32-50R/2          | 0.1     | 0.1   | 0.1   | 0.2    | -      | -      | -      |
| TP 32-80R/2          | 0.1     | 0.1   | 0.2   | 0.5    | -      | -      | -      |
| TP 32-90R/2          | 0.1     | 0.1   | 0.2   | 0.5    | -      | -      | -      |
| TP, TPD 32-60/2      | 0.1     | 0.1   | 0.2   | 1.0    | 1.5    | 3.2    | -      |
| TP, TPD 32-120/2     | 0.1     | 0.2   | 0.7   | 1.5    | 2.0    | 3.7    | -      |
| TP, TPD 32-150/2     | 0.1     | 0.3   | 0.8   | 1.6    | 2.1    | 3.8    | -      |
| TP, TPD 32-180/2     | 0.5     | 0.7   | 1.2   | 2.0    | 2.5    | 4.2    | -      |
| TP, TPD 32-230/2     | 0.7     | 0.9   | 1.4   | 2.2    | 2.7    | 4.4    | -      |
| TP, TPD 32-200/2     | 0.1     | 0.1   | 0.2   | 0.9    | 1.5    | 3.1    | -      |
| TP, TPD 32-250/2     | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.2    | -      |
| TP, TPD 32-320/2     | 0.1     | 0.1   | 0.6   | 1.3    | 1.9    | 3.5    | -      |
| TP, TPD 32-380/2     | 0.1     | 0.2   | 0.7   | 1.4    | 2.0    | 3.6    | -      |
| TP, TPD 32-460/2     | 0.1     | 0.2   | 0.7   | 1.4    | 1.9    | 3.6    | -      |
| TP, TPD 32-580/2     | 0.2     | 0.4   | 0.9   | 1.6    | 2.2    | 3.8    | -      |
| TP 40-50/2           | 0.1     | 0.1   | 0.1   | 0.3    | -      | -      | -      |
| TP 40-80/2           | 0.1     | 0.1   | 0.2   | 0.5    | -      | -      | -      |
| TP 40-90/2           | 0.1     | 0.1   | 0.2   | 0.5    | -      | -      | -      |
| TP, TPD 40-60/2      | 0.1     | 0.1   | 0.5   | 1.2    | 1.8    | 3.5    | -      |
| TP, TPD 40-120/2     | 0.1     | 0.1   | 0.4   | 1.2    | 1.7    | 3.4    | -      |
| TP 40-180/2          | 0.1     | 0.2   | 0.7   | 1.5    | 2.0    | 3.7    | -      |
| TP, TPD 40-190/2     | 0.1     | 0.3   | 0.8   | 1.6    | 2.1    | 3.8    | -      |
| TP, TPD 40-230/2     | 0.7     | 0.9   | 1.4   | 2.2    | 2.7    | 4.4    | -      |
| TP, TPD 40-270/2     | 0.7     | 0.9   | 1.4   | 2.2    | 2.7    | 4.4    | -      |
| TP, TPD 40-240/2     | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.3    | -      |
| TP, TPD 40-300/2     | 0.1     | 0.1   | 0.4   | 1.1    | 1.6    | 3.3    | -      |
| TP, TPD 40-360/2     | 0.2     | 0.4   | 0.9   | 1.6    | 2.1    | 3.8    | -      |
| TP, TPD 40-430/2     | 0.1     | 0.1   | 0.5   | 1.2    | 1.8    | 3.4    | -      |
| TP, TPD 40-530/2     | 0.1     | 0.1   | 0.6   | 1.3    | 1.9    | 3.5    | -      |
| TP, TPD 40-630/2     | 0.1     | 0.3   | 0.8   | 1.5    | 2.1    | 3.7    | -      |
| TP, TPD 50-60/2      | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.4    | -      |
| TP, TPD 50-120/2     | 0.1     | 0.2   | 0.7   | 1.5    | 2.0    | 3.7    | -      |
| TP, TPD 50-180/2     | 0.1     | 0.2   | 0.7   | 1.4    | 2.0    | 3.7    | -      |
| TP, TPD 50-160/2     | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.0    | -      |

| Pump type<br>(50 Hz) | p [bar] |       |       |        |        |        |        |
|----------------------|---------|-------|-------|--------|--------|--------|--------|
|                      | 20 °C   | 60 °C | 90 °C | 110 °C | 120 °C | 140 °C | 150 °C |
| TP, TPD 50-190/2     | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.0    | -      |
| TP, TPD 50-240/2     | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.0    | -      |
| TP, TPD 50-290/2     | 0.1     | 0.1   | 0.2   | 0.9    | 1.5    | 3.1    | -      |
| TP, TPD 50-360/2     | 0.1     | 0.1   | 0.2   | 1.0    | 1.5    | 3.1    | -      |
| TP, TPD 50-430/2     | 0.1     | 0.1   | 0.4   | 1.1    | 1.6    | 3.3    | -      |
| TP, TPD 50-420/2     | 0.1     | 0.1   | 0.3   | 1.1    | 1.6    | 3.2    | -      |
| TP, TPD 50-540/2     | 0.1     | 0.1   | 0.5   | 1.3    | 1.8    | 3.4    | -      |
| TP, TPD 50-630/2     | 0.1     | 0.1   | 0.6   | 1.4    | 1.9    | 3.6    | -      |
| TP, TPD 50-710/2     | 0.6     | 0.8   | 1.3   | 2.0    | 2.6    | 4.2    | -      |
| TP, TPD 50-830/2     | 0.5     | 0.7   | 1.2   | 2.0    | 2.5    | 4.1    | -      |
| TP, TPD 50-960/2     | 1.0     | 1.2   | 1.7   | 2.4    | 3.0    | 4.6    | -      |
| TP, TPD 65-60/2      | 0.1     | 0.3   | 0.8   | 1.5    | 2.1    | 3.8    | -      |
| TP, TPD 65-120/2     | 0.5     | 0.7   | 1.2   | 2.0    | 2.5    | 4.2    | -      |
| TP, TPD 65-180/2     | 0.3     | 0.5   | 1.0   | 1.8    | 2.3    | 4.0    | -      |
| TP, TPD 65-170/2     | 0.1     | 0.1   | 0.1   | 0.9    | 1.4    | 3.1    | -      |
| TP, TPD 65-210/2     | 0.1     | 0.1   | 0.2   | 0.9    | 1.5    | 3.1    | -      |
| TP, TPD 65-250/2     | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.2    | -      |
| TP, TPD 65-340/2     | 0.1     | 0.1   | 0.2   | 0.9    | 1.4    | 3.1    | -      |
| TP, TPD 65-410/2     | 0.1     | 0.1   | 0.2   | 0.9    | 1.4    | 3.1    | -      |
| TP, TPD 65-460/2     | 0.1     | 0.1   | 0.2   | 1.0    | 1.5    | 3.1    | -      |
| TP, TPD 65-550/2     | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.2    | -      |
| TP, TPD 65-660/2     | 0.1     | 0.1   | 0.4   | 1.1    | 1.6    | 3.3    | -      |
| TP, TPD 65-720/2     | 0.1     | 0.1   | 0.6   | 1.3    | 1.9    | 3.5    | -      |
| TP, TPD 65-930/2     | 0.6     | 0.8   | 1.3   | 2.0    | 2.6    | 4.2    | -      |
| TP, TPD 80-120/2     | 1.2     | 1.4   | 1.9   | 2.7    | 3.2    | 4.9    | -      |
| TP, TPD 80-140/2     | 0.1     | 0.2   | 0.7   | 1.4    | 1.9    | 3.6    | -      |
| TP, TPD 80-180/2     | 0.1     | 0.1   | 0.3   | 1.1    | 1.6    | 3.2    | -      |
| TP, TPD 80-210/2     | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.3    | -      |
| TP, TPD 80-240/2     | 0.1     | 0.1   | 0.5   | 1.3    | 1.8    | 3.4    | -      |
| TP, TPD 80-250/2     | 0.1     | 0.3   | 0.8   | 1.6    | 2.1    | 3.7    | -      |
| TP, TPD 80-330/2     | 0.1     | 0.2   | 0.7   | 1.4    | 2.0    | 3.6    | -      |
| TP, TPD 80-400/2     | 0.2     | 0.4   | 0.9   | 1.6    | 2.2    | 3.8    | -      |
| TP, TPD 80-520/2     | 0.1     | 0.1   | 0.6   | 1.4    | 1.9    | 3.5    | -      |
| TP, TPD 80-570/2     | 0.1     | 0.3   | 0.8   | 1.6    | 2.1    | 3.7    | -      |
| TP, TPD 80-700/2     | 0.6     | 0.8   | 1.3   | 2.1    | 2.6    | 4.2    | -      |
| TP, TPD 100-120/2    | 1.9     | 2.1   | 2.6   | 3.4    | 3.9    | 5.6    | -      |
| TP, TPD 100-160/2    | 0.1     | 0.1   | 0.6   | 1.3    | 1.9    | 3.5    | -      |
| TP, TPD 100-200/2    | 0.1     | 0.1   | 0.4   | 1.2    | 1.7    | 3.3    | -      |
| TP, TPD 100-240/2    | 0.1     | 0.1   | 0.5   | 1.3    | 1.8    | 3.4    | -      |
| TP, TPD 100-250/2    | 0.6     | 0.8   | 1.3   | 2.0    | 2.5    | 4.2    | -      |
| TP, TPD 100-310/2    | 0.6     | 0.8   | 1.3   | 2.0    | 2.6    | 4.2    | -      |



| Pump type<br>(50 Hz) | p [bar] |       |       |        |        |        |        |
|----------------------|---------|-------|-------|--------|--------|--------|--------|
|                      | 20 °C   | 60 °C | 90 °C | 110 °C | 120 °C | 140 °C | 150 °C |
| TP, TPD 100-360/2    | 0.6     | 0.8   | 1.3   | 2.0    | 2.5    | 4.2    | -      |
| TP, TPD 100-390/2    | 1.0     | 1.2   | 1.7   | 2.4    | 3.0    | 4.6    | -      |
| TP, TPD 100-480/2    | 1.5     | 1.7   | 2.2   | 2.9    | 3.5    | 5.1    | -      |
| TP 100-530/2         | 1.6     | 1.8   | 2.2   | 3.2    | 3.7    | 5.3    | 6.6    |
| TP 100-650/2         | 1.4     | 1.6   | 2     | 3      | 3.5    | 5.1    | 6.4    |
| TP 100-800/2         | 1.3     | 1.5   | 1.9   | 2.9    | 3.4    | 5      | 6.3    |
| TP 100-950/2         | 1.3     | 1.5   | 1.9   | 2.9    | 3.4    | 5      | 6.3    |
| TP 100-1040/2        | 1.2     | 1.4   | 1.8   | 2.8    | 3.3    | 4.9    | 6.2    |
| TP 100-1200/2        | 1.2     | 1.4   | 1.8   | 2.8    | 3.3    | 4.9    | 6.2    |
| TP 100-1410/2        | 1.2     | 1.4   | 1.8   | 2.8    | 3.3    | 4.9    | 6.2    |

**50 Hz, 4-pole**

| Pump type<br>(50 Hz) | p [bar] |       |       |        |        |        |        |
|----------------------|---------|-------|-------|--------|--------|--------|--------|
|                      | 20 °C   | 60 °C | 90 °C | 110 °C | 120 °C | 140 °C | 150 °C |
| TP, TPD 32-30/4      | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.1    | -      |
| TP, TPD 32-40/4      | 0.1     | 0.1   | 0.1   | 0.9    | 1.4    | 3.1    | -      |
| TP, TPD 32-60/4      | 0.1     | 0.1   | 0.3   | 1.1    | 1.6    | 3.3    | -      |
| TP, TPD 32-80/4      | 0.1     | 0.1   | 0.1   | 0.5    | 1.0    | 2.7    | -      |
| TP, TPD 32-100/4     | 0.1     | 0.1   | 0.1   | 0.5    | 1.1    | 2.7    | -      |
| TP, TPD 32-120/4     | 0.1     | 0.1   | 0.1   | 0.6    | 1.1    | 2.7    | -      |
| TP, TPD 40-30/4      | 0.1     | 0.1   | 0.2   | 0.9    | 1.5    | 3.2    | -      |
| TP 40-60/4           | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.1    | -      |
| TP, TPD 40-90/4      | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.3    | -      |
| TP, TPD 40-100/4     | 0.1     | 0.1   | 0.2   | 0.9    | 1.5    | 3.1    | -      |
| TP, TPD 40-110/4     | 0.1     | 0.1   | 0.1   | 0.6    | 1.2    | 2.8    | -      |
| TP, TPD 40-140/4     | 0.1     | 0.1   | 0.1   | 0.7    | 1.3    | 2.9    | -      |
| TP, TPD 50-30/4      | 0.1     | 0.1   | 0.1   | 0.9    | 1.4    | 3.1    | -      |
| TP, TPD 50-60/4      | 0.1     | 0.1   | 0.2   | 0.9    | 1.5    | 3.2    | -      |
| TP, TPD 50-90/4      | 0.1     | 0.1   | 0.1   | 0.6    | 1.1    | 2.8    | -      |
| TP, TPD 50-80/4      | 0.1     | 0.1   | 0.1   | 0.8    | 1.3    | 3.0    | -      |
| TP, TPD 50-120/4     | 0.1     | 0.1   | 0.1   | 0.7    | 1.3    | 2.9    | -      |
| TP, TPD 50-140/4     | 0.1     | 0.1   | 0.1   | 0.7    | 1.3    | 2.9    | -      |
| TP, TPD 50-190/4     | 0.1     | 0.1   | 0.1   | 0.9    | 1.4    | 3.0    | -      |
| TP, TPD 50-230/4     | 0.1     | 0.1   | 0.2   | 1.0    | 1.5    | 3.1    | -      |
| TP, TPD 65-30/4      | 0.1     | 0.2   | 0.7   | 1.5    | 2.0    | 3.7    | -      |
| TP, TPD 65-60/4      | 0.2     | 0.4   | 0.9   | 1.6    | 2.2    | 3.9    | -      |
| TP, TPD 65-90/4      | 0.1     | 0.1   | 0.1   | 0.6    | 1.1    | 2.7    | -      |
| TP, TPD 65-110/4     | 0.1     | 0.1   | 0.1   | 0.6    | 1.1    | 2.7    | -      |
| TP, TPD 65-130/4     | 0.1     | 0.1   | 0.1   | 0.6    | 1.1    | 2.8    | -      |
| TP, TPD 65-150/4     | 0.1     | 0.1   | 0.1   | 0.6    | 1.2    | 2.8    | -      |
| TP, TPD 65-170/4     | 0.1     | 0.1   | 0.1   | 0.6    | 1.2    | 2.8    | -      |

| Pump type<br>(50 Hz) | p [bar] |       |       |        |        |        |        |
|----------------------|---------|-------|-------|--------|--------|--------|--------|
|                      | 20 °C   | 60 °C | 90 °C | 110 °C | 120 °C | 140 °C | 150 °C |
| TP, TPD 65-240/4     | 0.1     | 0.1   | 0.1   | 0.8    | 1.3    | 2.9    | -      |
| TP, TPD 80-30/4      | 0.8     | 1.0   | 1.5   | 2.2    | 2.8    | 4.5    | -      |
| TP, TPD 80-60/4      | 0.8     | 1.0   | 1.5   | 2.3    | 2.8    | 4.5    | -      |
| TP, TPD 80-70/4      | 0.1     | 0.1   | 0.1   | 0.8    | 1.3    | 2.9    | -      |
| TP, TPD 80-90/4      | 0.1     | 0.1   | 0.1   | 0.7    | 1.2    | 2.8    | -      |
| TP, TPD 80-110/4     | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.0    | -      |
| TP, TPD 80-150/4     | 0.1     | 0.1   | 0.1   | 0.8    | 1.3    | 2.9    | -      |
| TP, TPD 80-170/4     | 0.1     | 0.1   | 0.2   | 1.0    | 1.5    | 3.1    | -      |
| TP, TPD 80-240/4     | 0.1     | 0.1   | 0.3   | 1.0    | 1.5    | 3.2    | -      |
| TP, TPD 80-270/4     | 0.1     | 0.1   | 0.2   | 0.9    | 1.5    | 3.1    | -      |
| TP, TPD 80-340/4     | 0.1     | 0.1   | 0.3   | 1.1    | 1.6    | 3.2    | -      |
| TP, TPD 100-30/4     | 0.8     | 1.0   | 1.5   | 2.2    | 2.8    | 4.5    | -      |
| TP, TPD 100-60/4     | 0.6     | 0.8   | 1.3   | 2.0    | 2.6    | 4.3    | -      |
| TP, TPD 100-70/4     | 0.1     | 0.1   | 0.1   | 0.8    | 1.3    | 3.0    | -      |
| TP, TPD 100-90/4     | 0.1     | 0.1   | 0.1   | 0.9    | 1.4    | 3.0    | -      |
| TP, TPD 100-110/4    | 0.1     | 0.1   | 0.2   | 1.0    | 1.5    | 3.1    | -      |
| TP, TPD 100-130/4    | 0.1     | 0.1   | 0.6   | 1.3    | 1.9    | 3.5    | -      |
| TP 100-140/4         | 0.2     | 0.4   | 0.8   | 1.8    | 2.3    | 3.9    | 5.2    |
| TP, TPD 100-170/4    | 0.3     | 0.5   | 1.0   | 1.7    | 2.3    | 3.9    | 5.2    |
| TP, TPD 100-200/4    | 0.1     | 0.1   | 0.5   | 1.2    | 1.8    | 3.4    | 4.7    |
| TP, TPD 100-250/4    | 0.1     | 0.2   | 0.7   | 1.4    | 2.0    | 3.6    | 4.9    |
| TP, TPD 100-330/4    | 0.3     | 0.5   | 1.0   | 1.7    | 2.3    | 3.9    | 5.2    |
| TP, TPD 100-370/4    | 0.3     | 0.5   | 1.0   | 1.7    | 2.3    | 3.9    | 5.2    |
| TP, TPD 100-410/4    | 0.5     | 0.7   | 1.2   | 1.9    | 2.5    | 4.1    | 5.4    |
| TP 125-60/4          | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.0    | -      |
| TP 125-80/4          | 0.1     | 0.1   | 0.1   | 0.9    | 1.4    | 3.1    | -      |
| TP 125-95/4          | 0.1     | 0.1   | 0.2   | 0.9    | 1.5    | 3.1    | -      |
| TP, TPD 125-110/4    | 0.1     | 0.1   | 0.1   | 0.9    | 1.4    | 3.0    | -      |
| TP, TPD 125-130/4    | 0.1     | 0.1   | 0.2   | 0.9    | 1.5    | 3.1    | -      |
| TP 125-150/4         | 0.2     | 0.4   | 0.8   | 1.8    | 2.3    | 3.9    | 5.2    |
| TP, TPD 125-160/4    | 0.1     | 0.1   | 0.2   | 1.0    | 1.5    | 3.1    | -      |
| TP, TPD 125-190/4    | 0.1     | 0.1   | 0.2   | 0.9    | 1.5    | 3.1    | 4.4    |
| TP, TPD 125-230/4    | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.2    | 4.5    |
| TP, TPD 125-300/4    | 0.1     | 0.1   | 0.2   | 0.9    | 1.5    | 3.1    | 4.4    |
| TP, TPD 125-340/4    | 0.1     | 0.1   | 0.3   | 1.0    | 1.5    | 3.2    | 4.5    |
| TP, TPD 125-400/4    | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.2    | 4.5    |
| TP 150-70/4          | 0.1     | 0.1   | 0.3   | 1.1    | 1.6    | 3.2    | -      |
| TP 150-110/4         | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.3    | -      |
| TP 150-155/4         | 0.1     | 0.1   | 0.5   | 1.2    | 1.8    | 3.4    | -      |
| TP 150-170/4         | 0.1     | 0.1   | 0.6   | 1.3    | 1.9    | 3.5    | -      |
| TP, TPD 150-130/4    | 0.1     | 0.1   | 0.4   | 1.1    | 1.6    | 3.3    | 4.6    |

| Pump type<br>(50 Hz) | p [bar] |       |       |        |        |        |        |
|----------------------|---------|-------|-------|--------|--------|--------|--------|
|                      | 20 °C   | 60 °C | 90 °C | 110 °C | 120 °C | 140 °C | 150 °C |
| TP, TPD 150-160/4    | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.3    | 4.6    |
| TP, TPD 150-200/4    | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.3    | 4.6    |
| TP, TPD 150-220/4    | 0.1     | 0.1   | 0.5   | 1.2    | 1.8    | 3.4    | 4.7    |
| TP, TPD 150-250/4    | 0.1     | 0.1   | 0.6   | 1.3    | 1.9    | 3.5    | 4.8    |
| TP 150-260/4         | 0.1     | 0.1   | 0.5   | 1.2    | 1.8    | 3.4    | 4.7    |
| TP 150-280/4         | 0.1     | 0.3   | 0.8   | 1.5    | 2.1    | 3.7    | 5.0    |
| TP 150-340/4         | 0.1     | 0.2   | 0.7   | 1.5    | 2.0    | 3.6    | 4.9    |
| TP 150-390/4         | 0.1     | 0.2   | 0.7   | 1.4    | 2.0    | 3.6    | 4.9    |
| TP 150-450/4         | 0.1     | 0.1   | 0.5   | 1.2    | 1.8    | 3.4    | 4.7    |
| TP 150-520/4         | 0.1     | 0.1   | 1.0   | 1.5    | 1.9    | 3.5    | 4.8    |
| TP 150-660/4         | 0.1     | 0.2   | 0.7   | 1.4    | 1.9    | 3.6    | 4.9    |
| TP 150-680/4         | 0.1     | 0.2   | 0.7   | 1.4    | 2.0    | 3.6    | -      |
| TP 200-50/4          | 0.3     | 0.4   | 0.9   | 1.7    | 2.2    | 3.8    | -      |
| TP 200-70/4          | 0.1     | 0.3   | 0.8   | 1.5    | 2.1    | 3.7    | -      |
| TP 200-90/4          | 0.1     | 0.2   | 0.7   | 1.4    | 2.0    | 3.6    | -      |
| TP 200-130/4         | 0.1     | 0.1   | 0.5   | 1.2    | 1.8    | 3.4    | -      |
| TP 200-150/4         | 0.1     | 0.1   | 0.4   | 1.2    | 1.7    | 3.3    | -      |
| TP 200-160/4         | 0.3     | 0.5   | 1.0   | 1.7    | 2.3    | 3.9    | 5.2    |
| TP 200-190/4         | 0.2     | 0.4   | 0.9   | 1.6    | 2.2    | 3.8    | 5.1    |
| TP 200-200/4         | 0.2     | 0.4   | 0.9   | 1.6    | 2.1    | 3.8    | 5.1    |
| TP 200-240/4         | 0.1     | 0.2   | 0.7   | 1.4    | 2.0    | 3.6    | 4.9    |
| TP 200-270/4         | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.3    | 4.6    |
| TP 200-290/4         | 0.1     | 0.1   | 0.6   | 1.3    | 1.9    | 3.5    | 4.8    |
| TP 200-320/4         | 0.1     | 0.1   | 0.5   | 1.2    | 1.8    | 3.4    | 4.7    |
| TP 200-330/4         | 0.1     | 0.1   | 0.3   | 1.1    | 1.6    | 3.2    | 4.5    |
| TP 200-360/4         | 0.1     | 0.1   | 0.3   | 1.1    | 1.6    | 3.2    | 4.5    |
| TP 200-400/4         | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.2    | 4.5    |
| TP 200-410/4         | 0.1     | 0.2   | 0.7   | 1.4    | 1.9    | 3.6    | 4.9    |
| TP 200-470/4         | 0.1     | 0.1   | 0.4   | 1.1    | 1.6    | 3.3    | 4.6    |
| TP 200-530/4         | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.3    | 4.6    |
| TP 200-590/4         | 0.1     | 0.2   | 0.7   | 1.4    | 2.0    | 3.6    | 4.9    |
| TP 200-660/4         | 0.2     | 0.4   | 0.9   | 1.7    | 2.2    | 3.8    | 5.1    |
| TP 250-280/4         | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.1    | -      |
| TP 250-310/4         | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.1    | -      |
| TP 250-390/4         | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.1    | -      |
| TP 300-190/4         | 0.5     | 0.7   | 1.1   | 2.1    | 2.6    | 4.2    | 5.5    |
| TP 300-220/4         | 0.3     | 0.5   | 0.9   | 1.9    | 2.4    | 4      | 5.3    |
| TP 300-250/4         | 0.1     | 0.3   | 0.7   | 1.7    | 2.2    | 3.8    | 5.1    |
| TP 300-290/4         | 0.5     | 0.7   | 1.1   | 2.1    | 2.6    | 4.2    | 5.5    |
| TP 300-390/4         | 0.5     | 0.7   | 1.1   | 2.1    | 2.6    | 4.2    | 5.5    |
| TP 300-420/4         | 0.5     | 0.7   | 1.1   | 2.1    | 2.6    | 4.2    | 5.5    |

| Pump type<br>(50 Hz) | p [bar] |       |       |        |        |        |        |
|----------------------|---------|-------|-------|--------|--------|--------|--------|
|                      | 20 °C   | 60 °C | 90 °C | 110 °C | 120 °C | 140 °C | 150 °C |
| TP 300-430/4         | 0.5     | 0.7   | 1.1   | 2.1    | 2.6    | 4.2    | 5.5    |
| TP 300-500/4         | 0.4     | 0.6   | 1     | 2      | 2.5    | 4.1    | 5.4    |
| TP 300-550/4         | 0.3     | 0.5   | 0.9   | 1.9    | 2.4    | 4      | 5.3    |
| TP 350-230/4         | 2.0     | 2.2   | 2.6   | 3.6    | 4.1    | 5.7    | 7.0    |
| TP 350-280/4         | 2.0     | 2.2   | 2.6   | 3.6    | 4.1    | 5.7    | 7.0    |
| TP 350-310/4         | 2.0     | 2.2   | 2.6   | 3.6    | 4.1    | 5.7    | 7.0    |
| TP 350-360/4         | 1.5     | 1.7   | 2.1   | 3.1    | 3.6    | 5.2    | 6.5    |
| TP 350-420/4         | 1.4     | 1.6   | 2.0   | 3.0    | 3.5    | 5.1    | 6.4    |
| TP 350-480/4         | 1.3     | 1.5   | 1.9   | 2.9    | 3.4    | 5.0    | 6.3    |
| TP 350-530/4         | 0.5     | 0.7   | 1.1   | 2.1    | 2.6    | 4.2    | 5.5    |
| TP 350-650/4         | 0.4     | 0.6   | 1.0   | 2.0    | 2.5    | 4.1    | 5.4    |
| TP 350-780/4         | 0.3     | 0.5   | 0.9   | 1.9    | 2.4    | 4.0    | 5.3    |

50 Hz, 6-pole

| Pump type<br>(50 Hz) | p [bar] |       |       |        |        |        |        |
|----------------------|---------|-------|-------|--------|--------|--------|--------|
|                      | 20 °C   | 60 °C | 90 °C | 110 °C | 120 °C | 140 °C | 150 °C |
| TP, TPD 125-60/6     | 0.1     | 0.1   | 0.1   | 0.7    | 1.2    | 2.8    | -      |
| TP, TPD 125-70/6     | 0.1     | 0.1   | 0.1   | 0.7    | 1.3    | 2.9    | -      |
| TP, TPD 125-80/6     | 0.1     | 0.1   | 0.1   | 0.7    | 1.2    | 2.9    | -      |
| TP, TPD 125-100/6    | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.0    | -      |
| TP, TPD 125-130/6    | 0.1     | 0.1   | 0.1   | 0.7    | 1.3    | 2.9    | -      |
| TP, TPD 125-160/6    | 0.1     | 0.1   | 0.1   | 0.7    | 1.3    | 2.9    | -      |
| TP, TPD 150-60/6     | 0.1     | 0.1   | 0.1   | 0.7    | 1.3    | 2.9    | -      |
| TP, TPD 150-70/6     | 0.1     | 0.1   | 0.1   | 0.7    | 1.3    | 2.9    | -      |
| TP, TPD 150-90/6     | 0.1     | 0.1   | 0.1   | 0.8    | 1.3    | 2.9    | -      |
| TP, TPD 150-110/6    | 0.1     | 0.1   | 0.1   | 0.8    | 1.3    | 3.0    | -      |

60 Hz, 2-pole

| Pump type<br>(60 Hz) | p [bar] |       |       |        |        |        |        |
|----------------------|---------|-------|-------|--------|--------|--------|--------|
|                      | 20 °C   | 60 °C | 90 °C | 110 °C | 120 °C | 140 °C | 150 °C |
| TP 32-80/2           | 0.4     | 0.4   | 0.4   | 1.2    | 1.7    | 3.4    | -      |
| TP 32-160/2          | 0.4     | 0.6   | 1.1   | 1.9    | 2.4    | 4.1    | -      |
| TP 32-220/2          | 0.7     | 0.9   | 1.4   | 2.2    | 2.7    | 4.4    | -      |
| TP 32-260/2          | 0.7     | 0.9   | 1.4   | 2.2    | 2.7    | 4.4    | -      |
| TP 32-330/2          | 0.7     | 0.9   | 1.4   | 2.2    | 2.7    | 4.4    | -      |
| TP, TPD 32-300/2     | 0.1     | 0.1   | 0.1   | 0.7    | 1.2    | 2.8    | -      |
| TP, TPD 32-360/2     | 0.1     | 0.1   | 0.1   | 0.7    | 1.2    | 2.8    | -      |
| TP, TPD 32-450/2     | 0.1     | 0.1   | 0.1   | 0.7    | 1.2    | 2.8    | -      |
| TP, TPD 32-550/2     | 0.1     | 0.1   | 0.1   | 0.7    | 1.2    | 2.9    | -      |
| TP, TPD 32-680/2     | 0.1     | 0.1   | 0.2   | 0.9    | 1.5    | 3.1    | -      |
| TP, TPD 32-820/2     | 0.5     | 0.7   | 1.2   | 1.9    | 2.5    | 4.1    | -      |

| Pump type<br>(60 Hz) | p [bar] |       |       |        |        |        |        |
|----------------------|---------|-------|-------|--------|--------|--------|--------|
|                      | 20 °C   | 60 °C | 90 °C | 110 °C | 120 °C | 140 °C | 150 °C |
| TP 40-80/2           | 0.1     | 0.3   | 0.8   | 1.6    | 2.1    | 3.8    | -      |
| TP 40-160/2          | 0.1     | 0.2   | 0.7   | 1.5    | 2.0    | 3.7    | -      |
| TP 40-240/2          | 0.4     | 0.6   | 1.1   | 1.9    | 2.4    | 4.1    | -      |
| TP 40-270/2          | 0.7     | 0.9   | 1.4   | 2.2    | 2.7    | 4.4    | -      |
| TP 40-330/2          | 0.7     | 0.9   | 1.4   | 2.2    | 2.7    | 4.4    | -      |
| TP 40-390/2          | 0.7     | 0.9   | 1.4   | 2.2    | 2.7    | 4.4    | -      |
| TP, TPD 40-400/2     | 0.1     | 0.1   | 0.1   | 0.9    | 1.4    | 3.1    | -      |
| TP, TPD 40-460/2     | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.2    | -      |
| TP, TPD 40-530/2     | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.3    | -      |
| TP, TPD 40-690/2     | 0.1     | 0.2   | 0.7   | 1.4    | 2.0    | 3.6    | -      |
| TP, TPD 40-820/2     | 0.1     | 0.3   | 0.8   | 1.6    | 2.1    | 3.7    | -      |
| TP, TPD 40-920/2     | 0.4     | 0.6   | 1.1   | 1.8    | 2.4    | 4.0    | -      |
| TP 50-80/2           | 0.1     | 0.1   | 0.6   | 1.4    | 1.9    | 3.6    | -      |
| TP 50-160/2          | 0.4     | 0.6   | 1.1   | 1.9    | 2.4    | 4.1    | -      |
| TP 50-240/2          | 0.3     | 0.5   | 1.0   | 1.8    | 2.3    | 4.0    | -      |
| TP, TPD 50-250/2     | 0.1     | 0.1   | 0.2   | 1.0    | 1.5    | 3.1    | -      |
| TP, TPD 50-300/2     | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.2    | -      |
| TP, TPD 50-350/2     | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.2    | -      |
| TP, TPD 50-410/2     | 0.1     | 0.1   | 0.4   | 1.1    | 1.6    | 3.3    | -      |
| TP, TPD 50-430/2     | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.3    | -      |
| TP, TPD 50-530/2     | 0.1     | 0.1   | 0.5   | 1.3    | 1.8    | 3.5    | -      |
| TP, TPD 50-640/2     | 0.1     | 0.1   | 0.6   | 1.4    | 1.9    | 3.5    | -      |
| TP, TPD 50-720/2     | 0.1     | 0.3   | 0.8   | 1.6    | 2.1    | 3.7    | -      |
| TP, TPD 50-790/2     | 0.5     | 0.7   | 1.2   | 1.9    | 2.5    | 4.1    | -      |
| TP, TPD 50-880/2     | 0.8     | 1.0   | 1.5   | 2.2    | 2.8    | 4.4    | -      |
| TP 50-1050/2         | 1.1     | 1.3   | 1.8   | 2.5    | 3.1    | 4.7    | -      |
| TP 65-80/2           | 0.6     | 0.8   | 1.3   | 2.1    | 2.6    | 4.3    | -      |
| TP 65-160/2          | 1.1     | 1.3   | 1.8   | 2.6    | 3.1    | 4.8    | -      |
| TP 65-240/2          | 0.9     | 1.1   | 1.6   | 2.4    | 2.9    | 4.6    | -      |
| TP, TPD 65-200/2     | 0.1     | 0.1   | 0.3   | 1.0    | 1.5    | 3.2    | -      |
| TP, TPD 65-250/2     | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.3    | -      |
| TP, TPD 65-340/2     | 0.1     | 0.1   | 0.2   | 1.0    | 1.5    | 3.1    | -      |
| TP, TPD 65-390/2     | 0.1     | 0.1   | 0.3   | 1.0    | 1.5    | 3.2    | -      |
| TP, TPD 65-480/2     | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.2    | -      |
| TP, TPD 65-540/2     | 0.1     | 0.1   | 0.3   | 1.1    | 1.6    | 3.2    | -      |
| TP, TPD 65-630/2     | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.3    | -      |
| TP, TPD 65-740/2     | 0.1     | 0.1   | 0.6   | 1.3    | 1.9    | 3.5    | -      |
| TP, TPD 65-910/2     | 0.1     | 0.2   | 0.7   | 1.5    | 2.0    | 3.6    | -      |
| TP, TPD 65-920/2     | 0.1     | 0.2   | 0.7   | 1.4    | 2.0    | 3.6    | -      |
| TP, TPD 65-1050/2    | 0.1     | 0.2   | 0.7   | 1.5    | 2.0    | 3.6    | -      |

| Pump type<br>(60 Hz) | p [bar] |       |       |        |        |        |        |
|----------------------|---------|-------|-------|--------|--------|--------|--------|
|                      | 20 °C   | 60 °C | 90 °C | 110 °C | 120 °C | 140 °C | 150 °C |
| TP 80-160/2          | 2.1     | 2.3   | 2.8   | 3.6    | 4.1    | 5.8    | -      |
| TP, TPD 80-200/2     | 0.5     | 0.7   | 1.2   | 1.9    | 2.5    | 4.1    | -      |
| TP, TPD 80-240/2     | 0.1     | 0.2   | 0.7   | 1.4    | 2.0    | 3.6    | -      |
| TP, TPD 80-290/2     | 0.1     | 0.3   | 0.8   | 1.5    | 2.1    | 3.7    | -      |
| TP, TPD 80-330/2     | 0.2     | 0.4   | 0.9   | 1.7    | 2.2    | 3.8    | -      |
| TP, TPD 80-400/2     | 0.6     | 0.8   | 1.3   | 2.1    | 2.6    | 4.2    | -      |
| TP, TPD 80-480/2     | 0.1     | 0.3   | 0.8   | 1.5    | 2.1    | 3.7    | -      |
| TP, TPD 80-530/2     | 0.2     | 0.4   | 0.9   | 1.6    | 2.1    | 3.8    | -      |
| TP, TPD 80-640/2     | 0.6     | 0.8   | 1.3   | 2.0    | 2.6    | 4.2    | -      |
| TP, TPD 80-750/2     | 0.6     | 0.8   | 1.3   | 2.0    | 2.6    | 4.2    | -      |
| TP, TPD 100-230/2    | 0.4     | 0.6   | 1.1   | 1.9    | 2.4    | 4.0    | -      |
| TP, TPD 100-300/2    | 0.2     | 0.4   | 0.9   | 1.6    | 2.2    | 3.8    | -      |
| TP, TPD 100-370/2    | 0.3     | 0.5   | 1.0   | 1.7    | 2.3    | 3.9    | -      |
| TP, TPD 100-350/2    | 0.9     | 1.1   | 1.6   | 2.3    | 2.9    | 4.5    | -      |
| TP, TPD 100-380/2    | 1.2     | 1.4   | 1.9   | 2.6    | 3.2    | 4.8    | -      |
| TP, TPD 100-530/2    | 1.7     | 1.9   | 2.4   | 3.2    | 3.7    | 5.3    | -      |
| TP, TPD 100-630/2    | 1.4     | 1.6   | 2.1   | 2.8    | 3.3    | 5.0    | -      |
| TP, TPD 100-700/2    | 3.0     | 3.2   | 3.7   | 4.4    | 5.0    | 6.6    | -      |
| TP 100-760/2         | 1.7     | 1.9   | 2.3   | 3.3    | 3.8    | 5.4    | 6.7    |
| TP 100-940/2         | 1.6     | 1.8   | 2.2   | 3.2    | 3.7    | 5.3    | 6.6    |
| TP 100-1040/2        | 1.6     | 1.8   | 2.2   | 3.2    | 3.7    | 5.3    | 6.6    |
| TP 100-1200/2        | 1.9     | 2.1   | 2.5   | 3.5    | 4      | 5.6    | 6.9    |
| TP 100-1360/2        | 1.8     | 2     | 2.4   | 3.4    | 3.9    | 5.5    | 6.8    |
| TP 100-1510/2        | 1.8     | 2     | 2.4   | 3.4    | 3.9    | 5.5    | 6.8    |

60 Hz, 4-pole

| Pump type<br>(60 Hz) | p [bar] |       |       |        |        |        |        |
|----------------------|---------|-------|-------|--------|--------|--------|--------|
|                      | 20 °C   | 60 °C | 90 °C | 110 °C | 120 °C | 140 °C | 150 °C |
| TP 32-40/4           | 0.1     | 0.1   | 0.1   | 0.9    | 1.4    | 3.1    | -      |
| TP 32-80/4           | 0.1     | 0.1   | 0.5   | 1.3    | 1.8    | 3.5    | -      |
| TP, TPD 32-120/4     | 0.1     | 0.1   | 0.1   | 0.7    | 1.3    | 2.9    | -      |
| TP, TPD 32-140/4     | 0.1     | 0.1   | 0.1   | 0.7    | 1.3    | 2.9    | -      |
| TP, TPD 32-190/4     | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.0    | -      |
| TP 40-40/4           | 0.1     | 0.1   | 0.3   | 1.1    | 1.6    | 3.3    | -      |
| TP 40-80/4           | 0.1     | 0.1   | 0.2   | 1.0    | 1.5    | 3.2    | -      |
| TP, TPD 40-110/4     | 0.1     | 0.1   | 0.1   | 0.6    | 1.2    | 2.8    | -      |
| TP, TPD 40-150/4     | 0.1     | 0.1   | 0.1   | 0.7    | 1.3    | 2.9    | -      |
| TP, TPD 40-180/4     | 0.1     | 0.1   | 0.1   | 0.7    | 1.3    | 2.9    | -      |
| TP, TPD 40-230/4     | 0.1     | 0.1   | 0.1   | 0.8    | 1.3    | 3.0    | -      |
| TP 50-40/4           | 0.1     | 0.1   | 0.3   | 1.1    | 1.6    | 3.3    | -      |
| TP 50-80/4           | 0.1     | 0.1   | 0.3   | 1.1    | 1.6    | 3.3    | -      |

| Pump type<br>(60 Hz) | p [bar] |       |       |        |        |        |        |
|----------------------|---------|-------|-------|--------|--------|--------|--------|
|                      | 20 °C   | 60 °C | 90 °C | 110 °C | 120 °C | 140 °C | 150 °C |
| TP, TPD 50-100/4     | 0.1     | 0.1   | 0.1   | 0.7    | 1.3    | 2.9    | -      |
| TP, TPD 50-115/4     | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.0    | -      |
| TP, TPD 50-130/4     | 0.1     | 0.1   | 0.1   | 0.9    | 1.4    | 3.0    | -      |
| TP, TPD 50-180/4     | 0.1     | 0.1   | 0.1   | 0.8    | 1.3    | 3.0    | -      |
| TP, TPD 50-240/4     | 0.1     | 0.1   | 0.2   | 0.9    | 1.5    | 3.1    | -      |
| TP, TPD 50-270/4     | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.2    | -      |
| TP, TPD 50-340/4     | 0.1     | 0.2   | 0.7   | 1.4    | 2.0    | 3.6    | -      |
| TP 65-40/4           | 0.4     | 0.6   | 1.1   | 1.9    | 2.4    | 4.1    | -      |
| TP 65-80/4           | 0.7     | 0.9   | 1.4   | 2.2    | 2.7    | 4.4    | -      |
| TP, TPD 65-130/4     | 0.1     | 0.1   | 0.1   | 0.6    | 1.2    | 2.8    | -      |
| TP, TPD 65-150/4     | 0.1     | 0.1   | 0.1   | 0.6    | 1.2    | 2.8    | -      |
| TP, TPD 65-190/4     | 0.1     | 0.1   | 0.1   | 0.6    | 1.2    | 2.8    | -      |
| TP, TPD 65-230/4     | 0.1     | 0.1   | 0.1   | 0.7    | 1.3    | 2.9    | -      |
| TP, TPD 65-310/4     | 0.1     | 0.1   | 0.1   | 0.7    | 1.3    | 2.9    | -      |
| TP, TPD 65-330/4     | 0.1     | 0.1   | 0.1   | 0.3    | 0.8    | 2.5    | -      |
| TP 80-40/4           | 1.5     | 1.7   | 2.2   | 3.0    | 3.5    | 5.2    | -      |
| TP 80-80/4           | 1.6     | 1.8   | 2.3   | 3.1    | 3.6    | 5.3    | -      |
| TP, TPD 80-110/4     | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.0    | -      |
| TP, TPD 80-150/4     | 0.1     | 0.1   | 0.1   | 0.8    | 1.3    | 2.9    | -      |
| TP, TPD 80-170/4     | 0.1     | 0.1   | 0.1   | 0.8    | 1.3    | 3.0    | -      |
| TP, TPD 80-230/4     | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.2    | -      |
| TP, TPD 80-280/4     | 0.1     | 0.1   | 0.2   | 1.0    | 1.5    | 3.1    | -      |
| TP, TPD 80-340/4     | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.2    | -      |
| TP, TPD 80-410/4     | 0.1     | 0.1   | 0.5   | 1.2    | 1.8    | 3.4    | -      |
| TP, TPD 80-460/4     | 0.1     | 0.1   | 0.6   | 1.3    | 1.9    | 3.5    | -      |
| TP, TPD 80-510/4     | 0.1     | 0.2   | 0.7   | 1.5    | 2.0    | 3.6    | -      |
| TP 100-40/4          | 1.4     | 1.6   | 2.1   | 2.9    | 3.4    | 5.1    | -      |
| TP 100-80/4          | 1.2     | 1.4   | 1.9   | 2.7    | 3.2    | 4.9    | -      |
| TP, TPD 100-100/4    | 0.1     | 0.1   | 0.2   | 0.9    | 1.5    | 3.1    | -      |
| TP, TPD 100-130/4    | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.2    | -      |
| TP, TPD 100-150/4    | 0.1     | 0.1   | 0.6   | 1.3    | 1.9    | 3.5    | -      |
| TP, TPD 100-170/4    | 0.1     | 0.1   | 0.6   | 1.3    | 1.9    | 3.5    | -      |
| TP, TPD 100-200/4    | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.3    | 4.6    |
| TP, TPD 100-240/4    | 0.1     | 0.1   | 0.6   | 1.3    | 1.9    | 3.5    | 4.8    |
| TP, TPD 100-260/4    | 0.6     | 0.8   | 1.3   | 2.1    | 2.7    | 4.3    | 5.6    |
| TP, TPD 100-290/4    | 0.5     | 0.7   | 1.2   | 2.0    | 2.5    | 4.1    | 5.4    |
| TP, TPD 100-340/4    | 0.6     | 0.8   | 1.3   | 2.0    | 2.6    | 4.2    | 5.5    |
| TP, TPD 100-350/4    | 0.2     | 0.4   | 0.9   | 1.7    | 2.3    | 3.9    | 5.2    |
| TP, TPD 100-390/4    | 0.7     | 0.9   | 1.4   | 2.1    | 2.7    | 4.3    | 5.6    |
| TP, TPD 100-470/4    | 0.9     | 1.1   | 1.6   | 2.3    | 2.9    | 4.5    | 5.8    |
| TP 100-560/4         | 0.1     | 0.3   | 0.7   | 1.7    | 2.2    | 3.8    | 5.1    |

| Pump type<br>(60 Hz) | p [bar] |       |       |        |        |        |        |
|----------------------|---------|-------|-------|--------|--------|--------|--------|
|                      | 20 °C   | 60 °C | 90 °C | 110 °C | 120 °C | 140 °C | 150 °C |
| TP 125-80/4          | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.0    | -      |
| TP 125-110/4         | 0.1     | 0.1   | 0.2   | 0.9    | 1.5    | 3.1    | -      |
| TP 125-135/4         | 0.1     | 0.1   | 0.3   | 1.1    | 1.6    | 3.3    | -      |
| TP, TPD 125-130/4    | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.2    | -      |
| TP, TPD 125-160/4    | 0.1     | 0.1   | 0.3   | 1.1    | 1.6    | 3.2    | -      |
| TP, TPD 125-200/4    | 0.1     | 0.1   | 0.2   | 0.9    | 1.4    | 3.1    | -      |
| TP, TPD 125-230/4    | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.3    | -      |
| TP 125-220/4         | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.3    | 4.5    |
| TP, TPD 125-280/4    | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.3    | 4.5    |
| TP, TPD 125-340/4    | 0.1     | 0.1   | 0.5   | 1.2    | 1.8    | 3.4    | 4.6    |
| TP, TPD 125-365/4    | 0.3     | 0.5   | 1.0   | 1.7    | 2.3    | 3.9    | 5.2    |
| TP, TPD 125-420/4    | 0.1     | 0.1   | 0.3   | 1.0    | 1.6    | 3.2    | 4.5    |
| TP, TPD 125-480/4    | 0.1     | 0.1   | 0.5   | 1.2    | 1.8    | 3.4    | 4.7    |
| TP 125-550/4         | 0.1     | 0.3   | 0.7   | 1.7    | 2.2    | 3.8    | 5.1    |
| TP 125-580/4         | 0.1     | 0.3   | 0.7   | 1.7    | 2.2    | 3.8    | 5.1    |
| TP 150-130/4         | 0.1     | 0.1   | 0.5   | 1.2    | 1.8    | 3.4    | -      |
| TP 150-160/4         | 0.1     | 0.2   | 0.7   | 1.5    | 2.0    | 3.6    | -      |
| TP 150-200/4         | 0.2     | 0.4   | 0.9   | 1.6    | 2.1    | 3.8    | -      |
| TP 150-220/4         | 0.3     | 0.5   | 1.0   | 1.7    | 2.3    | 3.9    | -      |
| TP, TPD 150-180/4    | 0.1     | 0.2   | 0.7   | 1.4    | 1.9    | 3.6    | 4.9    |
| TP, TPD 150-210/4    | 0.1     | 0.2   | 0.7   | 1.4    | 2.0    | 3.6    | 4.9    |
| TP, TPD 150-240/4    | 0.1     | 0.2   | 0.7   | 1.5    | 2.0    | 3.6    | 4.9    |
| TP, TPD 150-300/4    | 0.1     | 0.3   | 0.8   | 1.5    | 2.1    | 3.7    | 5.0    |
| TP, TPD 150-340/4    | 0.1     | 0.3   | 0.8   | 1.5    | 2.1    | 3.7    | 5.0    |
| TP 150-360/4         | 0.3     | 0.5   | 1.0   | 1.8    | 2.3    | 4.0    | 5.3    |
| TP 150-400/4         | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.1    | 4.4    |
| TP 150-440/4         | 0.1     | 0.1   | 0.4   | 1.1    | 1.7    | 3.3    | 4.6    |
| TP 150-480/4         | 0.1     | 0.1   | 0.5   | 1.3    | 1.8    | 3.4    | 4.7    |
| TP 150-610/4         | 0.1     | 0.2   | 0.7   | 1.4    | 2      | 3.6    | 4.9    |
| TP 150-700/4         | 0.1     | 0.3   | 0.8   | 1.5    | 2.1    | 3.7    | 5.0    |
| TP 150-810/4         | 0.3     | 0.4   | 0.9   | 1.7    | 2.2    | 3.8    | 5.1    |
| TP 150-960/4         | 0.4     | 0.6   | 1.1   | 1.8    | 2.3    | 3.8    | 5.1    |
| TP 200-80/4          | 0.9     | 1.1   | 1.6   | 2.3    | 2.9    | 4.5    | -      |
| TP 200-110/4         | 0.5     | 0.6   | 1.1   | 1.9    | 2.4    | 4.0    | -      |
| TP 200-140/4         | 0.3     | 0.5   | 1     | 1.7    | 2.3    | 3.9    | -      |
| TP 200-190/4         | 0.2     | 0.4   | 0.9   | 1.6    | 2.2    | 3.8    | -      |
| TP 200-210/4         | 0.1     | 0.2   | 0.7   | 1.4    | 2      | 3.6    | -      |
| TP 200-250/4         | 0.9     | 1.0   | 1.5   | 2.3    | 2.8    | 4.4    | 5.7    |
| TP 200-280/4         | 0.7     | 0.9   | 1.4   | 2.1    | 2.7    | 4.3    | 5.6    |
| TP 200-320/4         | 0.6     | 0.8   | 1.3   | 2.0    | 2.6    | 4.2    | 5.5    |
| TP 200-360/4         | 0.4     | 0.6   | 1.1   | 1.8    | 2.4    | 4.0    | 5.3    |



| Pump type<br>(60 Hz) | p [bar] |       |       |        |        |        |        |
|----------------------|---------|-------|-------|--------|--------|--------|--------|
|                      | 20 °C   | 60 °C | 90 °C | 110 °C | 120 °C | 140 °C | 150 °C |
| TP 200-390/4         | 0.3     | 0.5   | 1.0   | 1.7    | 2.2    | 3.9    | 5.2    |
| TP 200-400/4         | 0.1     | 0.1   | 0.6   | 1.3    | 1.9    | 3.6    | 4.9    |
| TP 200-430/4         | 0.1     | 0.1   | 0.6   | 1.4    | 1.9    | 3.6    | 4.9    |
| TP 200-440/4         | 0.1     | 0.2   | 0.7   | 1.5    | 2.0    | 3.7    | 5.0    |
| TP 200-490/4         | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.1    | 4.4    |
| TP 200-500/4         | 0.2     | 0.4   | 0.9   | 1.6    | 2.2    | 3.9    | 5.2    |
| TP 200-540/4         | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.1    | 4.4    |
| TP 200-600/4         | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.1    | 4.4    |
| TP 200-680/4         | 0.1     | 0.1   | 0.1   | 0.8    | 1.4    | 3.1    | 4.4    |
| TP 200-770/4         | 0.1     | 0.2   | 0.7   | 1.4    | 2.0    | 3.7    | 5.0    |
| TP 250-450/4         | 1.5     | 1.7   | 2.2   | 2.9    | 3.5    | 5.2    | -      |
| TP 250-530/4         | 1.5     | 1.7   | 2.2   | 2.9    | 3.5    | 5.2    | -      |
| TP 250-580/4         | 1.4     | 1.6   | 2.1   | 2.9    | 3.4    | 5.1    | -      |
| TP 300-230/4         | 0.8     | 1.0   | 1.4   | 2.4    | 2.9    | 4.5    | 5.8    |
| TP 300-270/4         | 0.7     | 0.9   | 1.3   | 2.3    | 2.8    | 4.4    | 5.7    |
| TP 300-360/4         | 0.7     | 0.9   | 1.3   | 2.3    | 2.8    | 4.4    | 5.7    |
| TP 300-370/4         | 0.8     | 1.0   | 1.4   | 2.4    | 2.9    | 4.5    | 5.8    |
| TP 300-440/4         | 0.8     | 1.0   | 1.4   | 2.4    | 2.9    | 4.5    | 5.8    |
| TP 300-550/4         | 0.8     | 1.0   | 1.4   | 2.4    | 2.9    | 4.5    | 5.8    |
| TP 300-630/4         | 0.8     | 1.0   | 1.4   | 2.4    | 2.9    | 4.5    | 5.8    |
| TP 300-640/4         | 0.7     | 0.9   | 1.3   | 2.3    | 2.8    | 4.4    | 5.7    |
| TP 300-750/4         | 0.7     | 0.9   | 1.3   | 2.3    | 2.8    | 4.4    | 5.7    |
| TP 350-280/4         | 2.0     | 2.2   | 2.6   | 3.6    | 4.1    | 5.7    | 7.0    |
| TP 350-330/4         | 1.9     | 2.1   | 2.5   | 3.5    | 4.0    | 5.6    | 6.9    |
| TP 350-390/4         | 1.9     | 2.1   | 2.5   | 3.5    | 4.0    | 5.6    | 6.9    |
| TP 350-440/4         | 1.9     | 2.1   | 2.5   | 3.5    | 4.0    | 5.6    | 6.9    |
| TP 350-450/4         | 2.0     | 2.2   | 2.6   | 3.6    | 4.1    | 5.7    | 7.0    |
| TP 350-540/4         | 2.0     | 2.2   | 2.6   | 3.6    | 4.1    | 5.7    | 7.0    |
| TP 350-680/4         | 2.0     | 2.2   | 2.6   | 3.6    | 4.1    | 5.7    | 7.0    |

Maximum sound pressure level

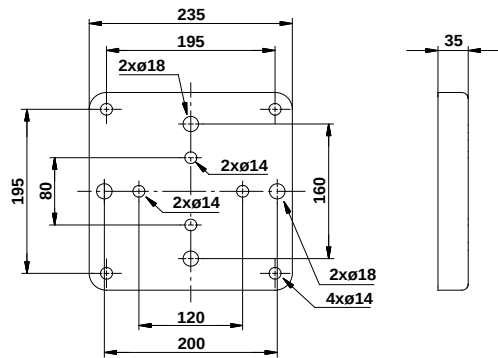
| Three-phase motors<br>[kW] | 50 Hz<br>[dB(A)] |        |        | 60 Hz<br>[dB(A)] |        |
|----------------------------|------------------|--------|--------|------------------|--------|
|                            | 2-pole           | 4-pole | 6-pole | 2-pole           | 4-pole |
| 0.12                       | < 70             | < 70   | -      | < 70             | < 70   |
| 0.18                       | < 70             | < 70   | -      | < 70             | < 70   |
| 0.25                       | 56               | 41     | -      | < 70             | 45     |
| 0.37                       | 56               | 45     | -      | 57               | 45     |
| 0.55                       | 57               | 42     | -      | 56               | 45     |
| 0.75                       | 53               | 59.5   | -      | 57               | 49     |
| 1.1                        | 53               | 49.5   | -      | 58               | 53     |
| 1.5                        | 58               | 50     | 47     | 64               | 53     |
| 2.2                        | 60               | 51     | 52     | 65               | 55     |
| 3.0                        | 59.5             | 53     | 63     | 53.5             | 55     |
| 4.0                        | 63               | 54     | 63     | 67.5             | 57     |
| 5.5                        | 62               | 50     | 63     | 68               | 62     |
| 7.5                        | 60               | 51     | 66     | 65               | 62     |
| 11.0                       | 60               | 53     | -      | 64.5             | 66     |
| 15.0                       | 60               | 66     | -      | 65               | 66     |
| 18.5                       | 60.5             | 63     | -      | 65.5             | 63     |
| 22.0                       | 65.5             | 63     | -      | 70.5             | 63     |
| 30.0                       | 70               | 65     | -      | 75               | 65     |
| 37.0                       | 71               | 66     | -      | 75               | 65     |
| 45.0                       | 67               | 66     | -      | 75               | 65     |
| 55.0                       | 72               | 67     | -      | 75               | 68     |
| 75.0                       | 74               | 70     | -      | 77               | 71     |
| 90.0                       | 73               | 70     | -      | 77               | 71     |
| 110.0                      | 76               | 70     | -      | 81               | 75     |
| 132.0                      | 76               | 70     | -      | 81               | 75     |
| 160.0                      | 76               | 70     | -      | 81               | 75     |
| 200.0                      | -                | 70     | -      | 81               | 75     |
| 250.0                      | -                | 73     | -      | 86               | 77     |
| 315.0                      | -                | 73     | -      | -                | 77     |
| 355.0                      | -                | 75     | -      | -                | -      |
| 400.0                      | -                | 75     | -      | -                | -      |
| 500.0                      | -                | 75     | -      | -                | -      |
| 560.0                      | -                | 78     | -      | -                | -      |
| 630.0                      | -                | 78     | -      | -                | -      |

## TP, TPE pumps from 11 kW and up suspended in the pipes

| Pump type         | PN 16 | PN 25 | P2<br>[kW] |  |  |  |  |  |  |  |  |
|-------------------|-------|-------|------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>50 Hz</b>      |       |       |            |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |
| TP, TPE 65-460/2  | ●     | -     | 11         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 65-550/2  | ●     | -     | 15         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 65-660/2  | ●     | -     | 18.5       |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 65-720/2  | ●     | -     | 22         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 80-330/2  | ●     | -     | 11         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 80-400/2  | ●     | -     | 15         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 80-520/2  | ●     | -     | 18.5       |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 80-570/2  | ●     | -     | 22         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 100-250/2 | ●     | -     | 11         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 100-310/2 | ●     | -     | 15         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 100-360/2 | ●     | -     | 18.5       |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 100-390/2 | ●     | -     | 22         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 80-340/4  | ●     | -     | 11         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 100-250/4 | ●     | ●     | 11         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 100-330/4 | ●     | ●     | 15         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 100-370/4 | ●     | ●     | 18.5       |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP 100-410/4      | ●     | ●     | 22         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 125-190/4 | ●     | ●     | 11         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 125-230/4 | ●     | ●     | 15         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 125-300/4 | ●     | ●     | 18.5       |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP 125-340/4      | ●     | ●     | 22         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 150-200/4 | ●     | ●     | 15         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 150-220/4 | ●     | ●     | 18.5       |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP 150-250/4      | ●     | ●     | 22         |                                                                                   | -                                                                                 |                                                                                   |                                                                                   |                                                                                   | ●                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 150-260/4 | -     | ●     | 18.5       |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP 150-280/4      | -     | ●     | 22         |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP 150-340/4      | -     | ●     | 30         |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP 150-390/4      | -     | ●     | 37         |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP 150-450/4      | -     | ●     | 45         |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP 150-520/4      | -     | ●     | 55         |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP 150-660/4      | -     | ●     | 75         |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 200-160/4 | -     | ●     | 15         |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP, TPE 200-190/4 | -     | ●     | 18.5       |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP 200-200/4      | -     | ●     | 22         |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP 200-240/4      | -     | ●     | 30         |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP 200-270/4      | -     | ●     | 45         |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP 200-320/4      | -     | ●     | 55         |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP 200-330/4      | -     | ●     | 37         |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP 200-360/4      | -     | ●     | 45         |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP 200-400/4      | -     | ●     | 55         |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP 200-410/4      | -     | ●     | 75         |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |
| TP 200-470/4      | -     | ●     | 75         |                                                                                   | ●                                                                                 |                                                                                   |                                                                                   |                                                                                   | -                                                                                 |                                                                                   |                                                                                    |

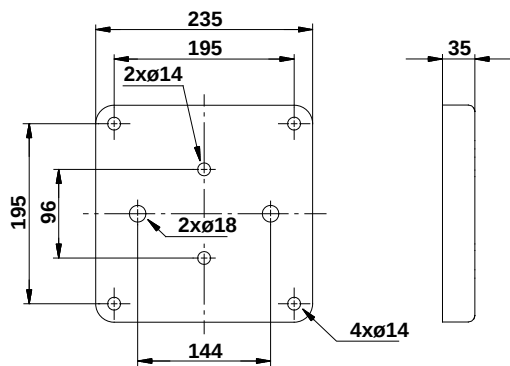
| Pump type         | PN 16 | PN 25 | P2<br>[kW] |  |  |  |  |  |  |  |  |
|-------------------|-------|-------|------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| TP 300-190/4      | -     | ●     | 30         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-220/4      | -     | ●     | 37         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-250/4      | -     | ●     | 45         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-290/4      | -     | ●     | 55         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-390/4      | -     | ●     | 75         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-420/4      | -     | ●     | 90         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-430/4      | -     | ●     | 110        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-500/4      | -     | ●     | 132        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-550/4      | -     | ●     | 160        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 350-230/4      | -     | ●     | 55         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 350-280/4      | -     | ●     | 75         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 350-310/4      | -     | ●     | 90         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 350-360/4      | -     | ●     | 110        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 350-420/4      | -     | ●     | 132        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 350-480/4      | -     | ●     | 160        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 350-530/4      | -     | ●     | 200        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 350-650/4      | -     | ●     | 250        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 350-780/4      | -     | ●     | 315        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| <b>60 Hz</b>      |       |       |            |                                                                                  |                                                                                  |                                                                                  |                                                                                  |                                                                                  |                                                                                  |                                                                                  |                                                                                   |
| TP, TPE 65-480/2  | ●     | -     | 11         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 65-540/2  | ●     | -     | 15         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 65-630/2  | ●     | -     | 18.5       |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 65-740/2  | ●     | -     | 22         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 80-330/2  | ●     | -     | 11         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 80-400/2  | ●     | -     | 15         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 80-480/2  | ●     | -     | 18.5       |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 80-530/2  | ●     | -     | 22         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 100-300/2 | ●     | -     | 11         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 100-370/2 | ●     | -     | 15         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 100-350/2 | ●     | -     | 18.5       |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 100-380/2 | ●     | -     | 22         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 80-340/4  | ●     | -     | 11         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 80-410/4  | ●     | -     | 15         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 80-460/4  | ●     | -     | 18.5       |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP 80-510/4       | ●     | -     | 22         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 100-240/4 | ●     | ●     | 11         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 100-260/4 | ●     | -     | 11         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 100-290/4 | ●     | ●     | 15         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 100-340/4 | ●     | ●     | 18.5       |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP 100-350/4      | ●     | -     | 22         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP 100-390/4      | ●     | ●     | 22         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP 100-470/4      | -     | ●     | 30         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 100-560/4      | -     | ●     | 37         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP, TPE 125-200/4 | ●     | -     | 11         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |

| Pump type         | PN 16 | PN 25 | P2<br>[kW] |  |  |  |  |  |  |  |  |
|-------------------|-------|-------|------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| TP, TPE 125-230/4 | ●     | -     | 15         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 125-220/4 | -     | ●     | 15         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 125-280/4 | ●     | ●     | 18.5       |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP 125-340/4      | ●     | ●     | 22         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP 125-365/4      | -     | ●     | 30         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 125-420/4      | -     | ●     | 30         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 125-480/4      | -     | ●     | 37         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 125-550/4      | -     | ●     | 45         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 125-580/4      | -     | ●     | 55         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP, TPE 150-180/4 | ●     | ●     | 15         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP, TPE 150-210/4 | ●     | ●     | 18.5       |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP 150-240/4      | ●     | ●     | 22         |                                                                                  | -                                                                                |                                                                                  |                                                                                  |                                                                                  | ●                                                                                |                                                                                  |                                                                                   |
| TP 150-300/4      | -     | ●     | 30         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 150-340/4      | -     | ●     | 37         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 150-360/4      | -     | ●     | 30         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 150-400/4      | -     | ●     | 37         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 150-440/4      | -     | ●     | 45         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 150-480/4      | -     | ●     | 55         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 150-610/4      | -     | ●     | 75         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 150-810/4      | -     | ●     | 110        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 150-960/4      | -     | ●     | 132        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 200-250/4      | -     | ●     | 30         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 200-400/4      | -     | ●     | 75         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 200-430/4      | -     | ●     | 55         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-230/4      | -     | ●     | 45         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-270/4      | -     | ●     | 55         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-360/4      | -     | ●     | 75         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-370/4      | -     | ●     | 90         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-440/4      | -     | ●     | 110        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-550/4      | -     | ●     | 132        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-630/4      | -     | ●     | 160        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-640/4      | -     | ●     | 200        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 300-750/4      | -     | ●     | 250        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 350-280/4      | -     | ●     | 90         |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 350-330/4      | -     | ●     | 110        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 350-390/4      | -     | ●     | 132        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 350-440/4      | -     | ●     | 160        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 350-450/4      | -     | ●     | 160        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 350-540/4      | -     | ●     | 200        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |
| TP 350-680/4      | -     | ●     | 250        |                                                                                  | ●                                                                                |                                                                                  |                                                                                  |                                                                                  | -                                                                                |                                                                                  |                                                                                   |



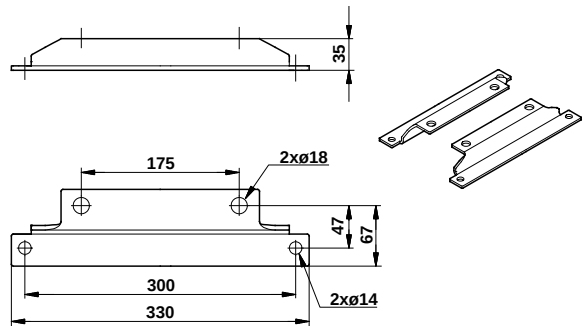
TM00 9835 0497

Fig. 1



TM00 3755 5097

Fig. 2



TM02 5336 2602

Fig. 3

## Declaration of conformity

### GB: EC/EU declaration of conformity

We, Grundfos, declare under our sole responsibility that the products TP, TPD, to which the declaration below relates, are in conformity with the Council Directives listed below on the approximation of the laws of the EC/EU member states.

**Note:** There are two sets of Council Directives and standards listed below. One set applies until and including 19th April 2016. The other set applies from 20th April 2016 and onwards.

These Directives and standards apply until and including 19th April 2016:

- Machinery Directive (2006/42/EC).  
Standards used: EN 809:1998 + A1:2009.
- Ecodesign Directive (2009/125/EC).  
Electric motors:  
Commission Regulation No 640/2009.  
Applies only to three-phase Grundfos motors marked IE2 or IE3. See motor nameplate.  
Standard used: EN 60034-30:2009.
- Ecodesign Directive (2009/125/EC).  
Water pumps:  
Commission Regulation No 547/2012.  
Applies only to water pumps marked with the minimum efficiency index MEI. See pump nameplate.
- ATEX Directive (94/9/EC).  
Applies only to products with the ATEX mark on the nameplate.  
Declaration of conformity and installation and operating instructions of the motor are enclosed.

Standards used: EN 13463-1:2009 and EN 13463-5:2011.

### CN: 欧共体 / 欧盟符合性声明

我们，格兰富，在我们的全权责任下声明，产品 TP、TPD，即该合格证所指之产品，符合欧共体 / 欧盟使其成员国法律趋于一致的以下理事会指令。

**注意：**有两套理事会指令和标准，如下所示。一套将于 2016 年 4 月 19 日失效。另一套将自 2016 年 4 月 20 日起生效。

These Directives and standards apply from 20th April 2016 and onwards:

- Machinery Directive (2006/42/EC).  
Standards used: EN 809:1998 + A1:2009.
- Ecodesign Directive (2009/125/EC).  
Electric motors:  
Commission Regulation No 640/2009.  
Applies only to three-phase Grundfos motors marked IE2 or IE3. See motor nameplate.  
Standard used: EN 60034-30:2009.
- Ecodesign Directive (2009/125/EC).  
Water pumps:  
Commission Regulation No 547/2012.  
Applies only to water pumps marked with the minimum efficiency index MEI. See pump nameplate.
- ATEX Directive (2014/34/EU).  
Applies only to products with the ATEX mark on the nameplate.  
Declaration of conformity and installation and operating instructions of the motor are enclosed.

Standards used: EN 13463-1:2009 and EN 13463-5:2011.

This EC/EU declaration of conformity is only valid when published as part of the Grundfos installation and operating instructions (publication number 96653082 0516).

Suzhou, Jiang Su Province, PRC 28th September 2015



Mette Bjerrekær  
Director, D&E China  
Grundfos (China) Holding Ltd.  
No.72, Qingqiu Road  
Suzhou, Jiangsu 215126, China

Person empowered to sign the EC/EU declaration of conformity.

Svend Aage Kaas  
Director  
Grundfos Holding A/S  
Poul Due Jensens Vej 7  
8850 Bjerringbro, Denmark

Person authorised to compile the technical file.

**Argentina**

Bombas GRUNDFOS de Argentina S.A.  
Ruta Panamericana km. 37.500 Centro  
Industrial Garin  
1619 Garin Pcia. de B.A.  
Phone: +54-3327 414 444  
Telefax: +54-3327 45 3190

**Australia**

GRUNDFOS Pumps Pty. Ltd.  
P.O. Box 2040  
Regency Park  
South Australia 5942  
Phone: +61-8-8461-4611  
Telefax: +61-8-8340 0155

**Austria**

GRUNDFOS Pumpen Vertrieb  
Ges.m.b.H.  
Grundfosstraße 2  
A-5082 Grödig/Salzburg  
Tel.: +43-6246-883-0  
Telefax: +43-6246-883-30

**Belgium**

N.V. GRUNDFOS Bellux S.A.  
Boomsesteenweg 81-83  
B-2630 Aartselaar  
Tél.: +32-3-870 7300  
Télécopie: +32-3-870 7301

**Belarus**

Представительство ГРУНДФОС в  
Минске  
220125, Минск  
ул. Шафарнянская, 11, оф. 56, БЦ  
«Порт»  
Тел.: +7 (375 17) 286 39 72/73  
Факс: +7 (375 17) 286 39 71  
E-mail: minsk@grundfos.com

**Bosna and Herzegovina**

GRUNDFOS Sarajevo  
Zmaja od Bosne 7-7A,  
BH-71000 Sarajevo  
Phone: +387 33 592 480  
Telefax: +387 33 590 465  
www.ba.grundfos.com  
e-mail: grundfos@bih.net.ba

**Brazil**

BOMBAS GRUNDFOS DO BRASIL  
Av. Humberto de Alencar Castelo  
Branco, 630  
CEP 09850 - 300  
São Bernardo do Campo - SP  
Phone: +55-11 4393 5533  
Telefax: +55-11 4343 5015

**Bulgaria**

Grundfos Bulgaria EOOD  
Slatina District  
Iztochna Tangenta street no. 100  
BG - 1592 Sofia  
Tel. +359 2 49 22 200  
Fax. +359 2 49 22 201  
email: bulgaria@grundfos.bg

**Canada**

GRUNDFOS Canada Inc.  
2941 Brighton Road  
Oakville, Ontario  
L6H 6C9  
Phone: +1-905 829 9533  
Telefax: +1-905 829 9512

**China**

GRUNDFOS Pumps (Shanghai) Co. Ltd.  
10F The Hub, No. 33 Suhong Road  
Minhang District  
Shanghai 201106  
PRC  
Phone: +86 21 612 252 22  
Telefax: +86 21 612 253 33

**Croatia**

GRUNDFOS CROATIA d.o.o.  
Buzinski prilaz 38, Buzin  
HR-10010 Zagreb  
Phone: +385 1 6595 400  
Telefax: +385 1 6595 499  
www.hr.grundfos.com

**Czech Republic**

GRUNDFOS s.r.o.  
Čajkovského 21  
779 00 Olomouc  
Phone: +420-585-716 111  
Telefax: +420-585-716 299

**Denmark**

GRUNDFOS DK A/S  
Martin Bachs Vej 3  
DK-8850 Bjerringbro  
Tlf.: +45-87 50 50 50  
Telefax: +45-87 50 51 51  
E-mail: info\_GDK@grundfos.com  
www.grundfos.com/DK

**Estonia**

GRUNDFOS Pumps Eesti OÜ  
Peterburi tee 92G  
11415 Tallinn  
Tel: + 372 606 1690  
Fax: + 372 606 1691

**Finland**

OY GRUNDFOS Pumput AB  
Trukkikuja 1  
FI-01360 Vantaa  
Phone: +358-(0) 207 889 500  
Telefax: +358-(0) 207 889 550

**France**

Pompes GRUNDFOS Distribution S.A.  
Parc d'Activités de Chesnes  
57, rue de Malacombe  
F-38290 St. Quentin Fallavier (Lyon)  
Tél.: +33-4 74 82 15 15  
Télécopie: +33-4 74 94 10 51

**Germany**

GRUNDFOS GMBH  
Schlüterstr. 33  
40699 Erkrath  
Tel.: +49-(0) 211 929 69-0  
Telefax: +49-(0) 211 929 69-3799  
e-mail: infoservice@grundfos.de  
Service in Deutschland:  
e-mail: kundendienst@grundfos.de

**Greece**

GRUNDFOS Hellas A.E.B.E.  
20th km. Athinon-Markopoulou Av.  
P.O. Box 71  
GR-19002 Peania  
Phone: +0030-210-66 83 400  
Telefax: +0030-210-66 46 273

**Hong Kong**

GRUNDFOS Pumps (Hong Kong) Ltd.  
Unit 1, Ground floor  
Siu Wai Industrial Centre  
29-33 Wing Hong Street &  
68 King Lam Street, Cheung Sha Wan  
Kowloon  
Phone: +852-27861706 / 27861741  
Telefax: +852-27858664

**Hungary**

GRUNDFOS Hungária Kft.  
Park u. 8  
H-2045 Törökbálint,  
Phone: +36-23 511 110  
Telefax: +36-23 511 111

**India**

GRUNDFOS Pumps India Private  
Limited  
118 Old Mahabalipuram Road  
Thoraipakkam  
Chennai 600 096  
Phone: +91-44 2496 6800

**Indonesia**

PT. GRUNDFOS POMPA  
Graha Intirub Lt. 2 & 3  
Jln. Cililitan Besar No.454. Makassar,  
Jakarta Timur  
ID-Jakarta 13650  
Phone: +62 21-469-51900  
Telefax: +62 21-460 6910 / 460 6901

**Ireland**

GRUNDFOS (Ireland) Ltd.  
Unit A, Merrywell Business Park  
Ballymount Road Lower  
Dublin 12  
Phone: +353-1-4089 800  
Telefax: +353-1-4089 830

**Italy**

GRUNDFOS Pompe Italia S.r.l.  
Via Gran Sasso 4  
I-20060 Truccazzano (Milano)  
Tel.: +39-02-95838112  
Telefax: +39-02-95309290 / 95838461

**Japan**

GRUNDFOS Pumps K.K.  
Gotanda Metalion Bldg., 5F,  
5-21-15, Higashi-gotanda  
Shiagawa-ku, Tokyo  
141-0022 Japan  
Phone: +81 35 448 1391  
Telefax: +81 35 448 9619

**Korea**

GRUNDFOS Pumps Korea Ltd.  
6th Floor, Aju Building 679-5  
Yeoksam-dong, Kangnam-ku, 135-916  
Seoul, Korea  
Phone: +82-2-5317 600  
Telefax: +82-2-5633 725

**Latvia**

SIA GRUNDFOS Pumps Latvia  
Deglava biznesa centrs  
Augusta Deglava iela 60, LV-1035, Rīga,  
Tālr.: + 371 714 9640, 7 149 641  
Fakss: + 371 914 9646

**Lithuania**

GRUNDFOS Pumps UAB  
Smolensko g. 6  
LT-03201 Vilnius  
Tel: + 370 52 395 430  
Fax: + 370 52 395 431



**Malaysia**

GRUNDFOS Pumps Sdn. Bhd.  
7 Jalan Peguam U1/25  
Glenmarie Industrial Park  
40150 Shah Alam  
Selangor  
Phone: +60-3-5569 2922  
Telefax: +60-3-5569 2866

**Mexico**

Bombas GRUNDFOS de México S.A. de C.V.  
Boulevard TLC No. 15  
Parque Industrial Stiva Aeropuerto  
Apodaca, N.L. 66600  
Phone: +52-81-8144 4000  
Telefax: +52-81-8144 4010

**Netherlands**

GRUNDFOS Netherlands  
Veluwezoom 35  
1326 AE Almere  
Postbus 22015  
1302 CA ALMERE  
Tel.: +31-88-478 6336  
Telefax: +31-88-478 6332  
E-mail: info\_gnl@grundfos.com

**New Zealand**

GRUNDFOS Pumps NZ Ltd.  
17 Beatrice Tinsley Crescent  
North Harbour Industrial Estate  
Albany, Auckland  
Phone: +64-9-415 3240  
Telefax: +64-9-415 3250

**Norway**

GRUNDFOS Pumper A/S  
Strømsveien 344  
Postboks 235, Leirdal  
N-1011 Oslo  
Tlf.: +47-22 90 47 00  
Telefax: +47-22 32 21 50

**Poland**

GRUNDFOS Pompy Sp. z o.o.  
ul. Klonowa 23  
Baranowo k. Poznań  
PL-62-081 Przeźmierowo  
Tel: (+48-61) 650 13 00  
Fax: (+48-61) 650 13 50

**Portugal**

Bombas GRUNDFOS Portugal, S.A.  
Rua Calvet de Magalhães, 241  
Apartado 1079  
P-2770-153 Paço de Arcos  
Tel.: +351-21-440 76 00  
Telefax: +351-21-440 76 90

**Romania**

GRUNDFOS Pompe România SRL  
Bd. Biruintei, nr 103  
Pantelimon county Ilfov  
Phone: +40 21 200 4100  
Telefax: +40 21 200 4101  
E-mail: romania@grundfos.ro

**Russia**

ООО Грундфос Россия  
109544, г. Москва, ул. Школьная,  
39-41, стр. 1  
Тел. (+7) 495 564-88-00 (495)  
737-30-00  
Факс (+7) 495 564 88 11  
E-mail grundfos.moscow@grundfos.com

**Serbia**

Grundfos Srbija d.o.o.  
Omladinskih brigada 90b  
11070 Novi Beograd  
Phone: +381 11 2258 740  
Telefax: +381 11 2281 769  
www.rs.grundfos.com

**Singapore**

GRUNDFOS (Singapore) Pte. Ltd.  
25 Jalan Tukang  
Singapore 619264  
Phone: +65-6681 9688  
Telefax: +65-6681 9689

**Slovakia**

GRUNDFOS s.r.o.  
Prievozská 4D  
821 09 BRATISLAVA  
Phona: +421 2 5020 1426  
sk.grundfos.com

**Slovenia**

GRUNDFOS LJUBLJANA, d.o.o.  
Leskoškova 9e, 1122 Ljubljana  
Phone: +386 (0) 1 568 06 10  
Telefax: +386 (0) 1 568 06 19  
E-mail: tehnika-si@grundfos.com

**South Africa**

GRUNDFOS (PTY) LTD  
Corner Mountjoy and George Allen  
Roads  
Wilbart Ext. 2  
Bedfordview 2008  
Phone: (+27) 11 579 4800  
Fax: (+27) 11 455 6066  
E-mail: lsmart@grundfos.com

**Spain**

Bombas GRUNDFOS España S.A.  
Camino de la Fuenteclilla, s/n  
E-28110 Algete (Madrid)  
Tel.: +34-91-848 8800  
Telefax: +34-91-628 0465

**Sweden**

GRUNDFOS AB  
Box 333 (Lunnagårdsgatan 6)  
431 24 Mölndal  
Tel.: +46 31 332 23 000  
Telefax: +46 31 331 94 60

**Switzerland**

GRUNDFOS Pumpen AG  
Bruggacherstrasse 10  
CH-8117 Fällanden/ZH  
Tel.: +41-44-806 8111  
Telefax: +41-44-806 8115

**Taiwan**

GRUNDFOS Pumps (Taiwan) Ltd.  
7 Floor, 219 Min-Chuan Road  
Taichung, Taiwan, R.O.C.  
Phone: +886-4-2305 0868  
Telefax: +886-4-2305 0878

**Thailand**

GRUNDFOS (Thailand) Ltd.  
92 Chaloem Phrakiat Rama 9 Road,  
Dokmai, Pravej, Bangkok 10250  
Phone: +66-2-725 8999  
Telefax: +66-2-725 8998

**Turkey**

GRUNDFOS POMPA San. ve Tic. Ltd. Sti.  
Gebze Organize Sanayi Bölgesi  
İhsan dede Caddesi,  
2. yol 200. Sokak No. 204  
41490 Gebze/ Kocaeli  
Phone: +90 - 262-679 7979  
Telefax: +90 - 262-679 7905  
E-mail: satis@grundfos.com

**Ukraine**

Бізнес Центр Європа  
Столичне шосе, 103  
м. Київ, 03131, Україна  
Телефон: (+38 044) 237 04 00  
Факс.: (+38 044) 237 04 01  
E-mail: ukraine@grundfos.com

**United Arab Emirates**

GRUNDFOS Gulf Distribution  
P.O. Box 16768  
Jebel Ali Free Zone  
Dubai  
Phone: +971 4 8815 166  
Telefax: +971 4 8815 136

**United Kingdom**

GRUNDFOS Pumps Ltd.  
Grovebury Road  
Leighton Buzzard/Beds. LU7 4TL  
Phone: +44-1525-850000  
Telefax: +44-1525-850011

**U.S.A.**

GRUNDFOS Pumps Corporation  
17100 West 118th Terrace  
Olathe, Kansas 66061  
Phone: +1-913-227-3400  
Telefax: +1-913-227-3500

**Uzbekistan**

Grundfos Tashkent, Uzbekistan The Representative Office of Grundfos Kazakhstan in Uzbekistan  
38a, Oybek street, Tashkent  
Телефон: (+998) 71 150 3290 / 71 150 3291  
Факс: (+998) 71 150 3292

Addresses Revised 25.01.2016

|               |
|---------------|
| 96653082 0516 |
| ECM: 1183385  |

The name Grundfos, the Grundfos logo, and be think innovate are registered trademarks owned by Grundfos Holding A/S or Grundfos A/S, Denmark. All rights reserved worldwide. © Copyright Grundfos Holding A/S